

GPU Acceleration of
the PROMPI Fortran Code
using OpenACC (OpenMP is next)

Prof Raphael HIRSCHI

Keele University, UK &

Klaus Tschira Guest Professor 2025 at HITS



Scientific Context: Predicting the evolution, fate and nucleosynthesis of massive stars in 1 and 3D

Recording on the HITS YouTube channel: <https://youtu.be/ziA2QvbtTKQ>



Raphael Hirschi

Professor of
Astrophysics at Keele ...



SHEN @ Kele: E. Whitehead, K. Goodman (previously C. Chambers), Vishnu Varma

Ex Keele PhD students: F. Rizzuti, LJA. Scott, E. Kaiser, J. den Hartogh, A. Cristini, S. Jones, M. Bennett
in collaboration with:

BH-GW: K. Belczynski (NCAC, PL), J. Groh, E. Farrel (TCDublin, IE)

GVA code: G. Meynet, A. Maeder, C. Georgy, S. Ekström, P. Eggenberger, S. Martinet, A. Choplin (ULB) and C. Chiappini (IAP, D)

VMS: N. Yusof, H. Ismail, H. Kassim (UM, KL, Malaysia), P. Crowther (Sheffield), O. Schnurr (IAP)

Nucleo: F.-K. Thielemann, U. Frischknecht, T. Rauscher (Basel, CH) N. Nishimura (Kyoto, JP)

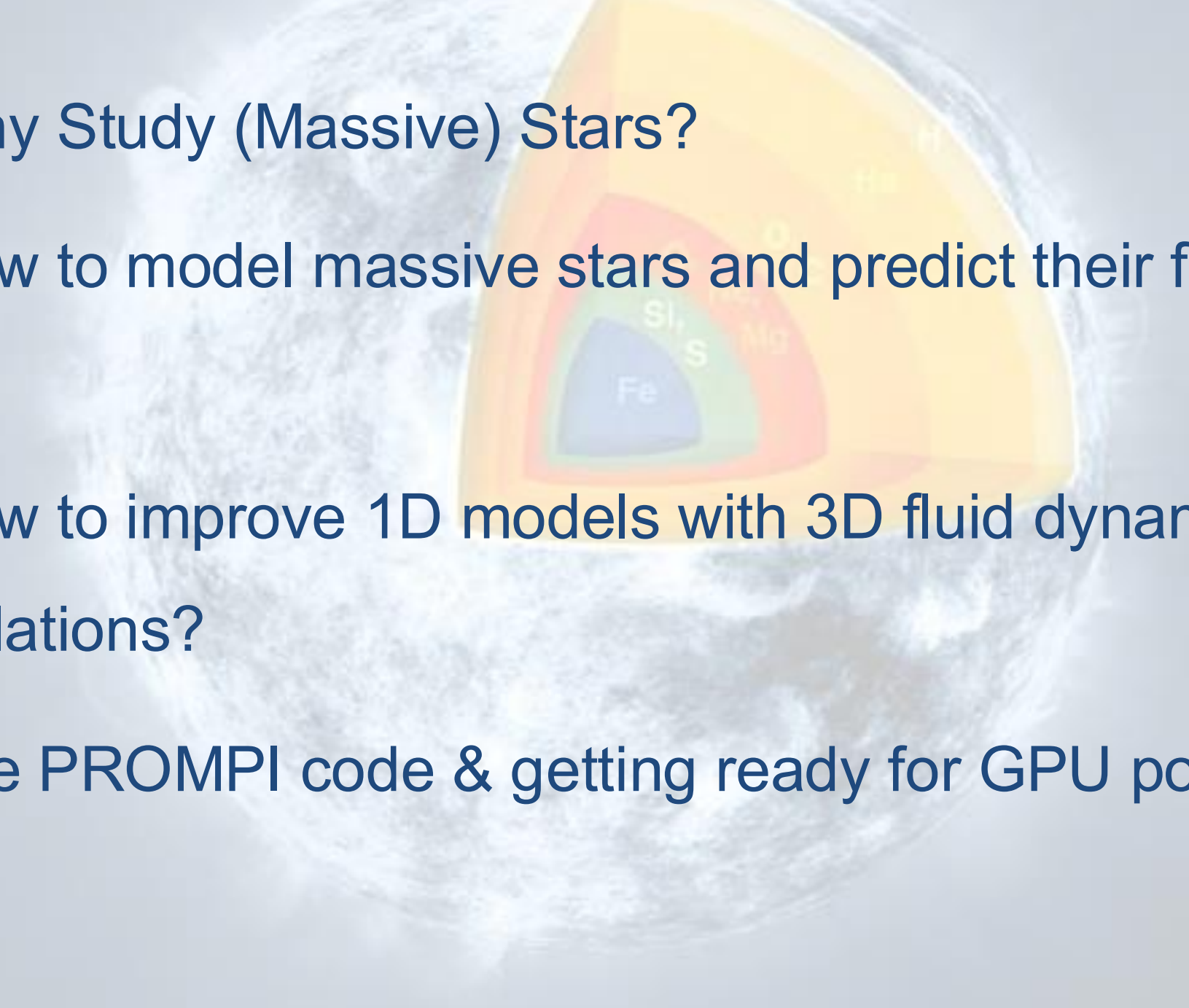
NUGRID: F. Herwig (UVIC, Canada), M. Pignatari (Konkoly), U. Battino (Hull), C. Fryer (LANL), Laird (York),

Wan Aishah Arun (U Edinburgh)...

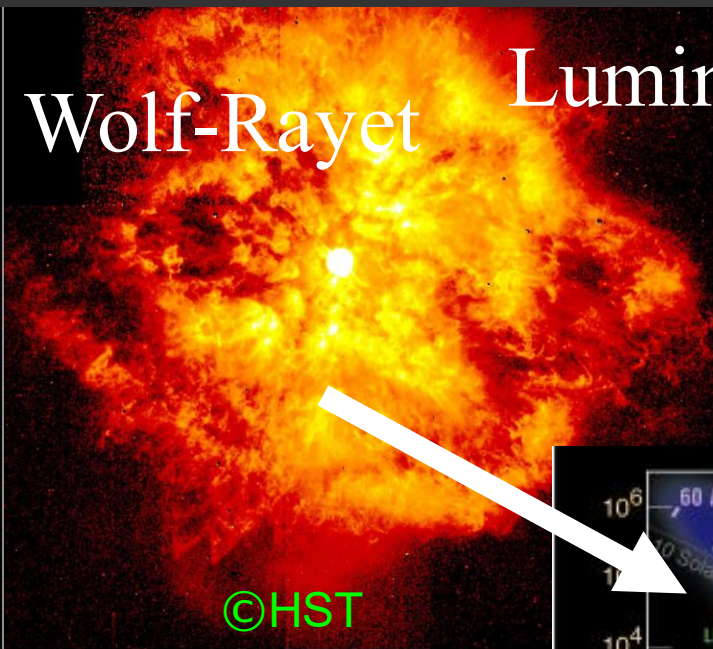
SNe: K. Nomoto (IPMU, J), C. Fröhlich (NCSU)

HYDRO: C. Meakin, D. Arnett (UArizona), C. Georgy (GVA), M. Mocak, F. Roepke, L. Horst (HITS, D), P. Edelmann (LANL)

Plan

- 
- Why Study (Massive) Stars?
 - How to model massive stars and predict their fate in 1D?
 - How to improve 1D models with 3D fluid dynamics simulations?
 - The PROMPI code & getting ready for GPU porting

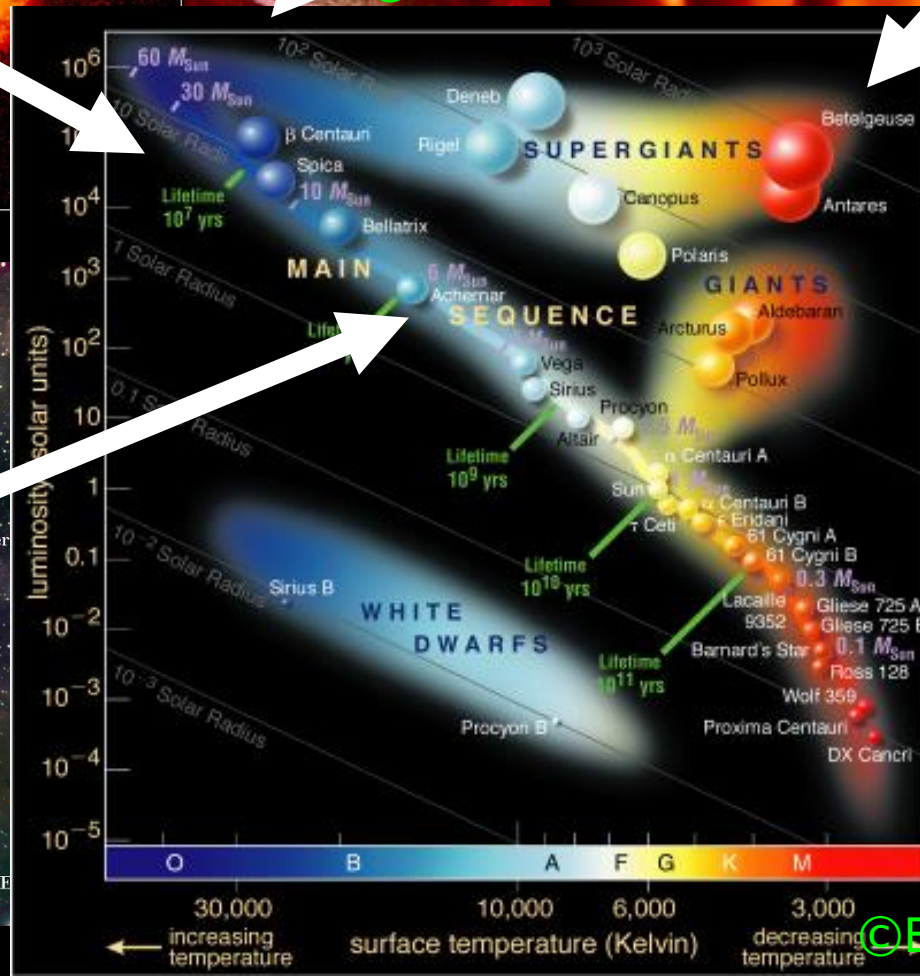
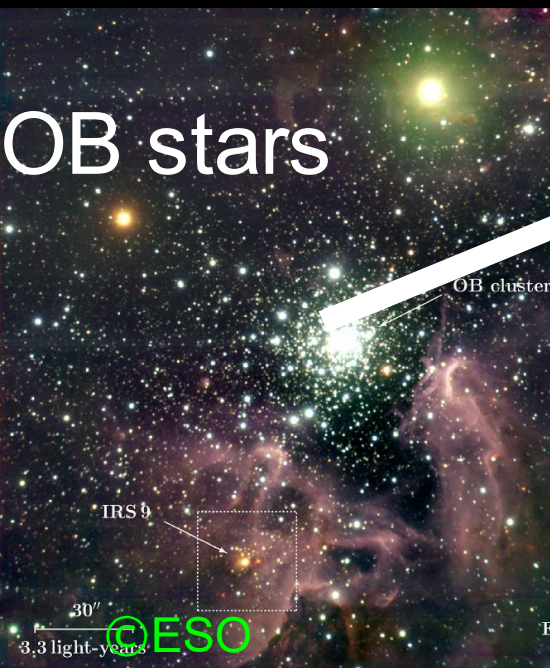
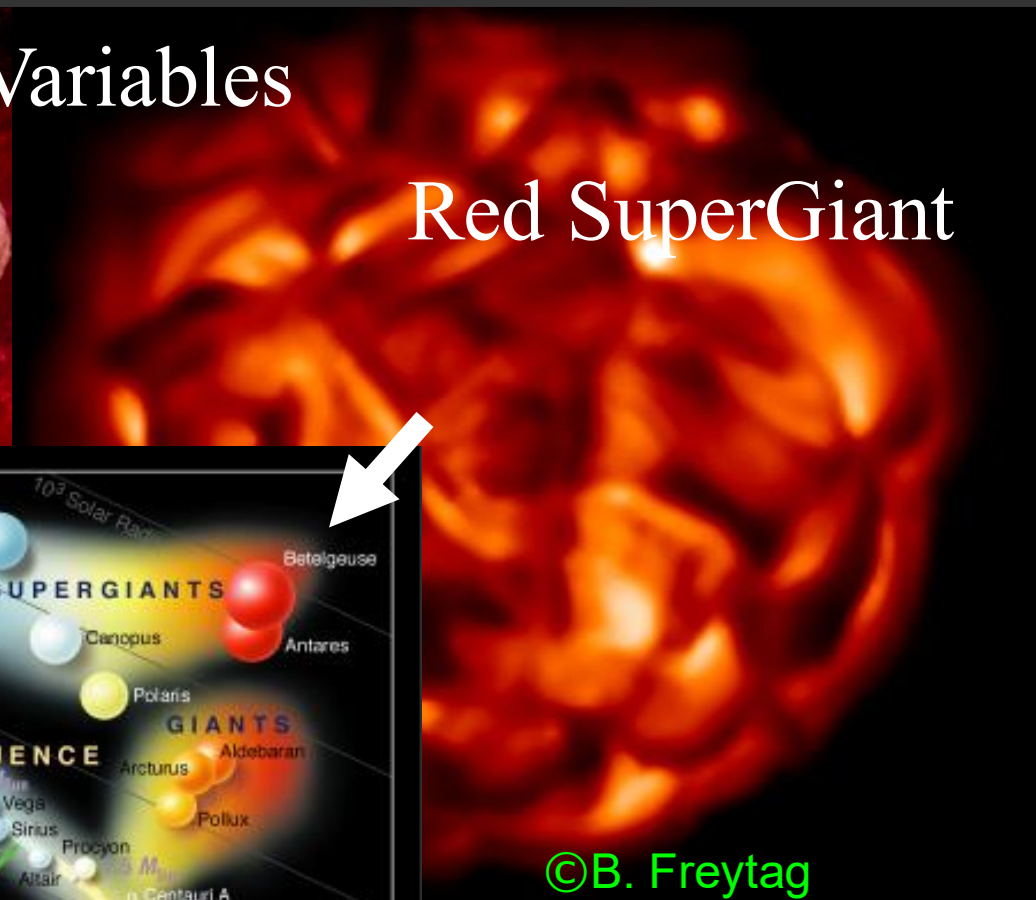
Importance as **Stellar Objects**



Luminous Blue Variables



Red SuperGiant



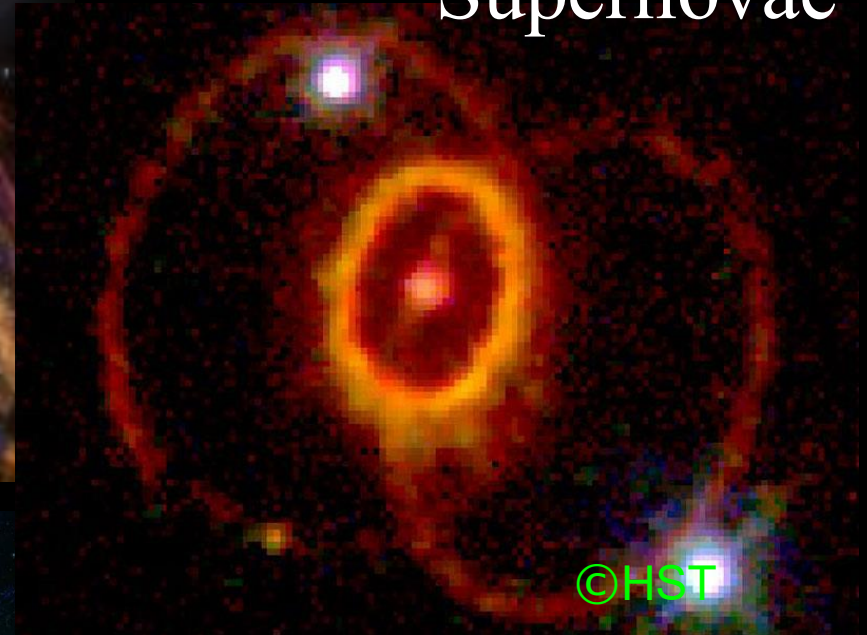
Importance as Progenitors

©NASA

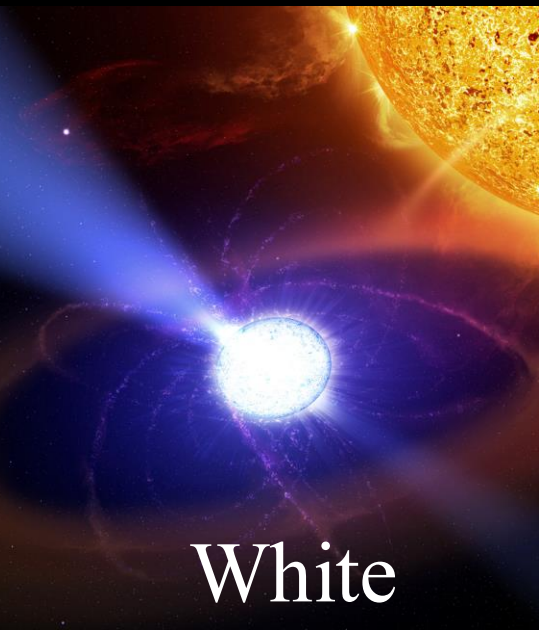


GRBs

Supernovae



©HST



White
Dwarfs



©STScI

Neutron
Stars



©UCLA

Black
Holes

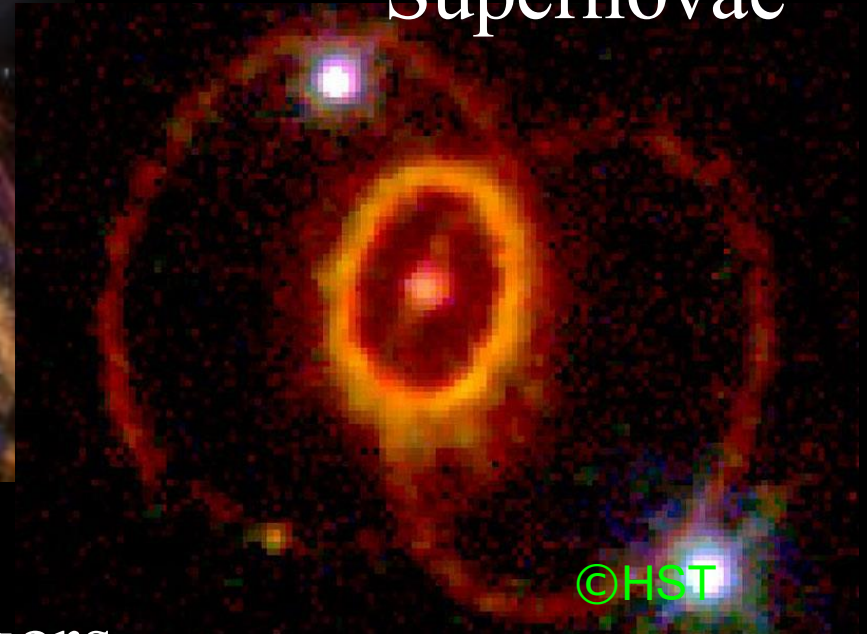
Stellar Explosions

©NASA



GRBs

Supernovae



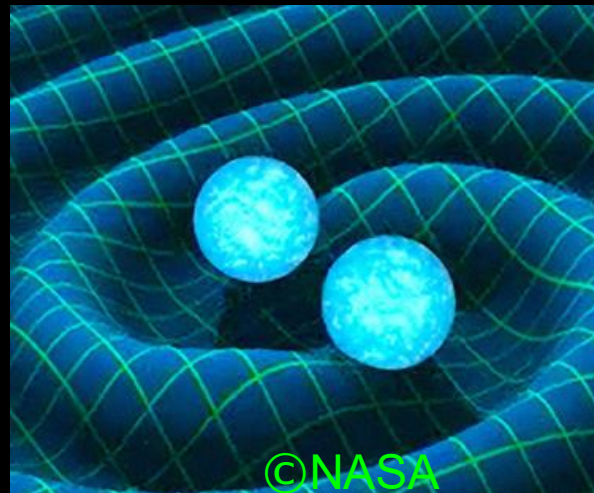
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Neutron
Stars

©STScI



GWs ← mergers



©NASA

Black Holes



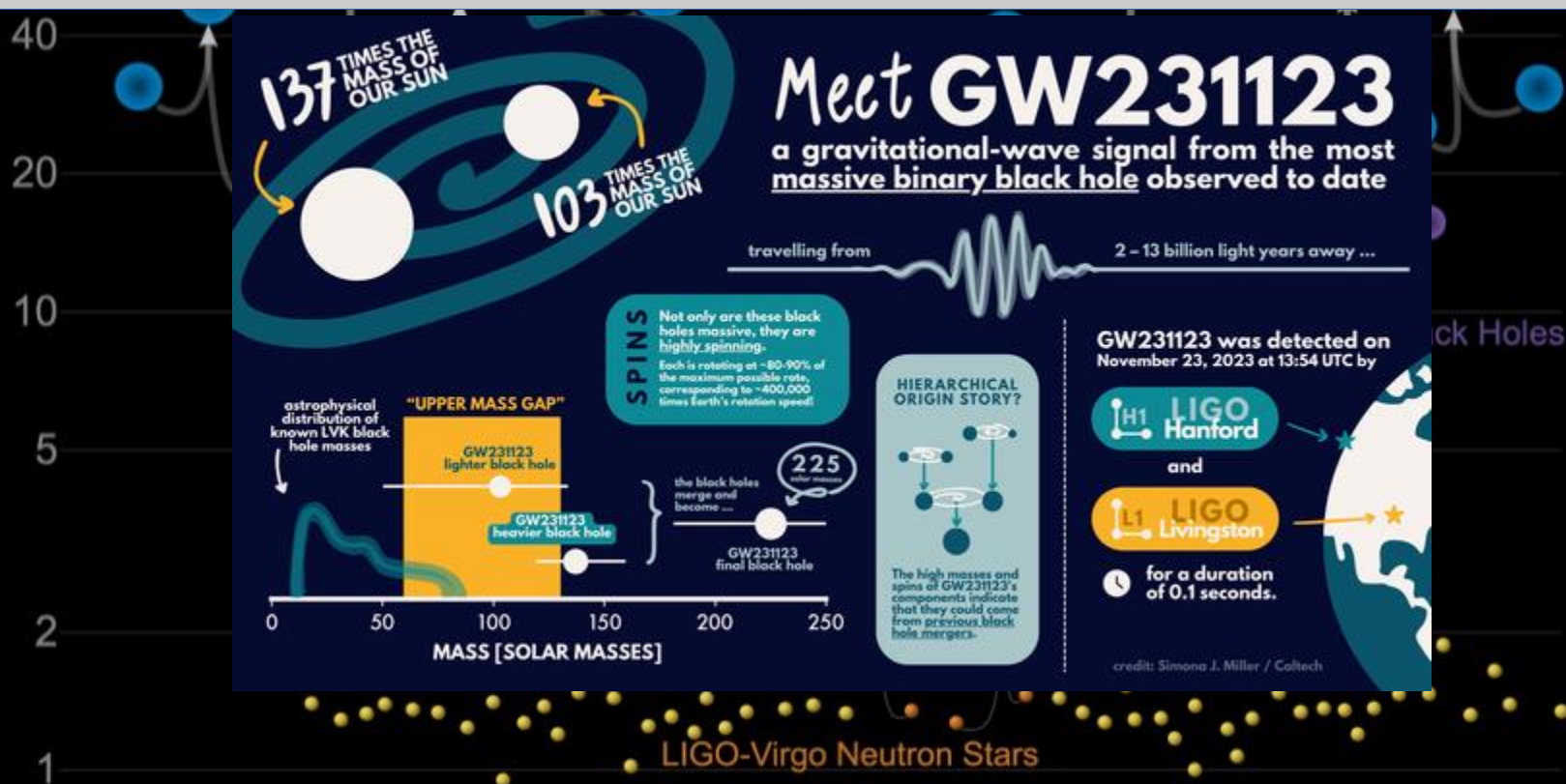
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Are there BH Mass Gaps?

Masses in the Stellar Graveyard *in Solar Masses*

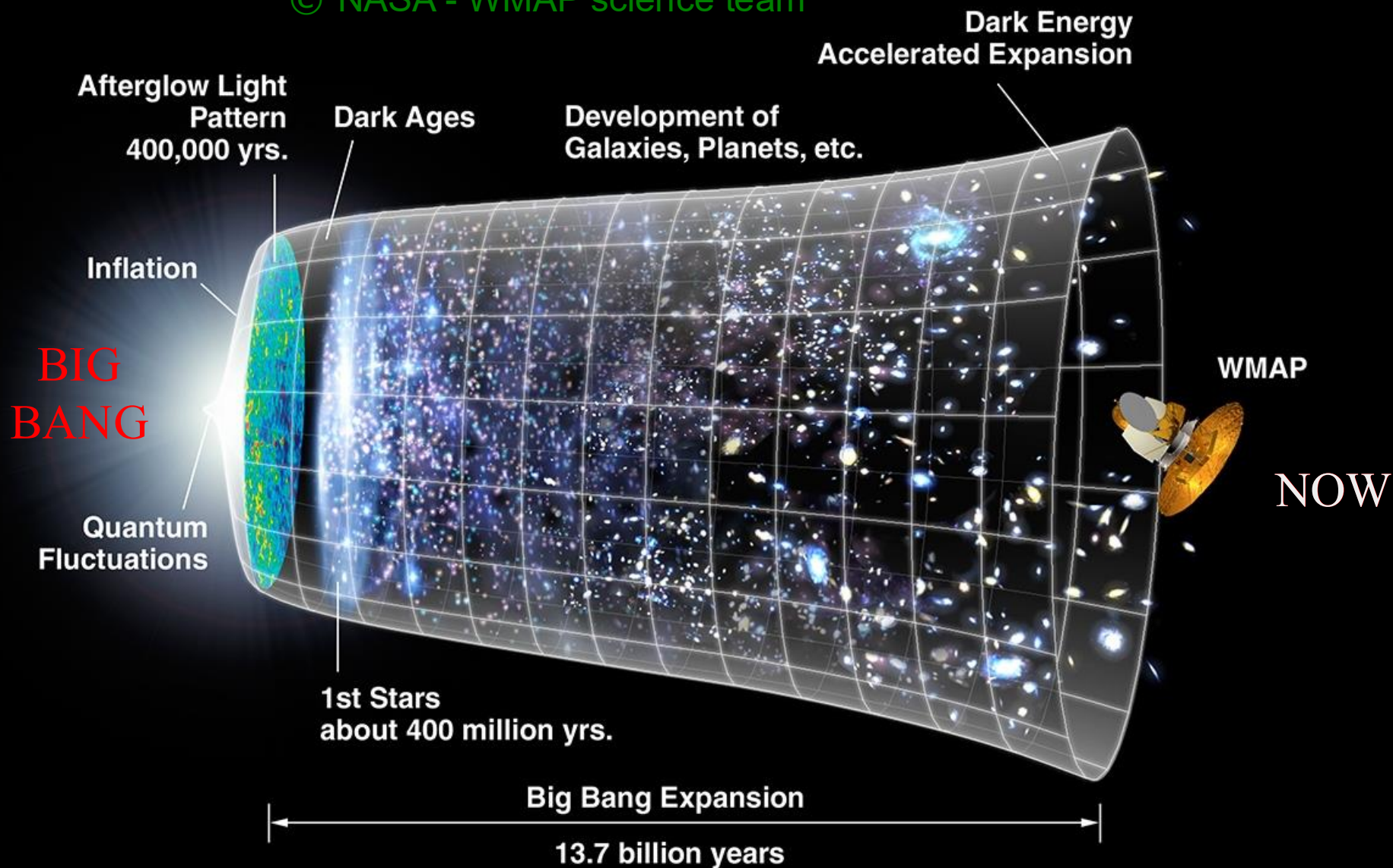
160

Is there a BH mass gap here, $50 M_{\odot} < M_{\text{BH}} < \sim 150 M_{\odot}$ (aka PISN-mass gap)?



Massive Stars' Role in Universe

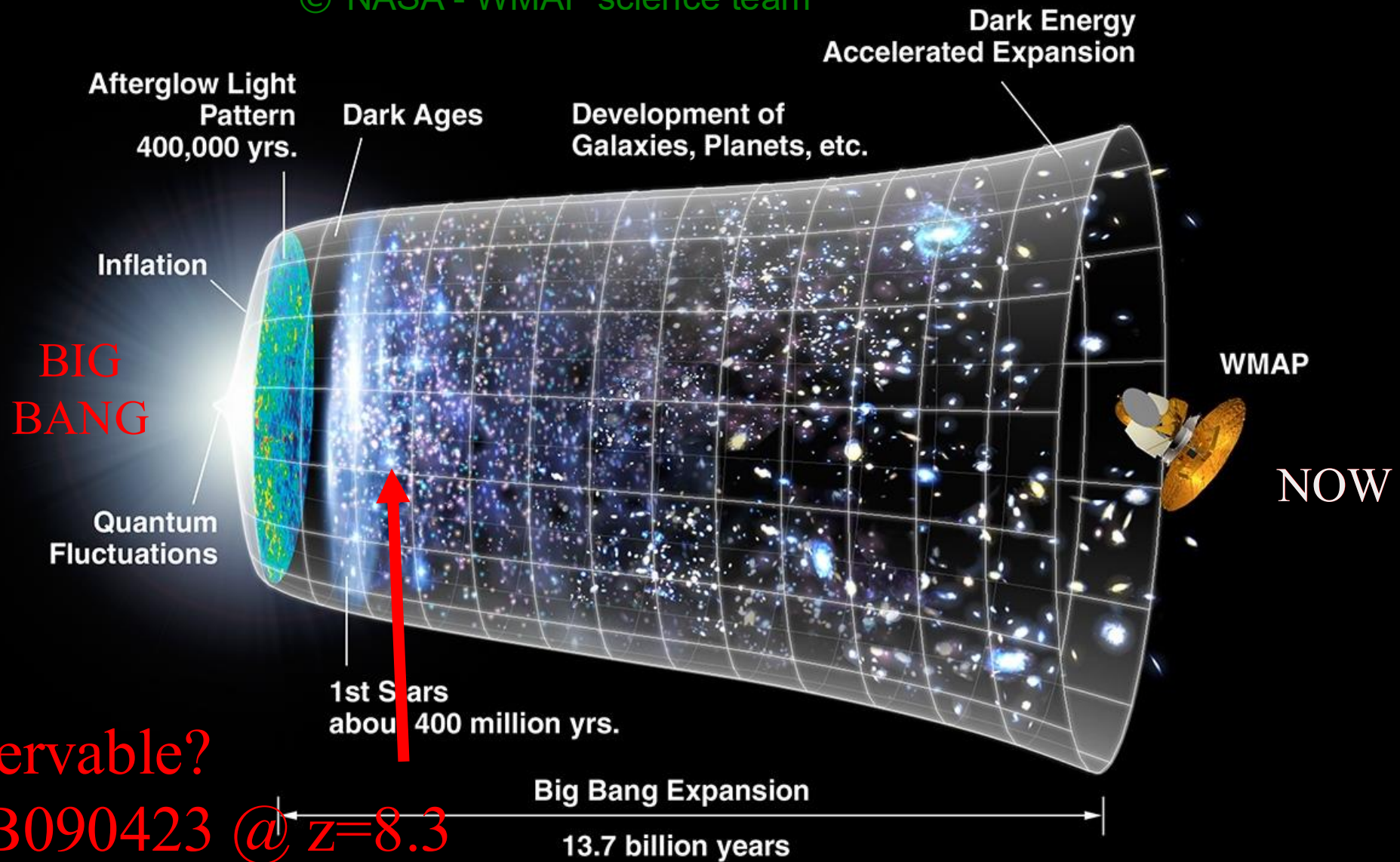
© NASA - WMAP science team



- Radiative, kinetic & chemical feedback
- Short-lived

First Stellar Generations

© NASA - WMAP science team



- Observable?

- GRB090423 @ $z=8.3$

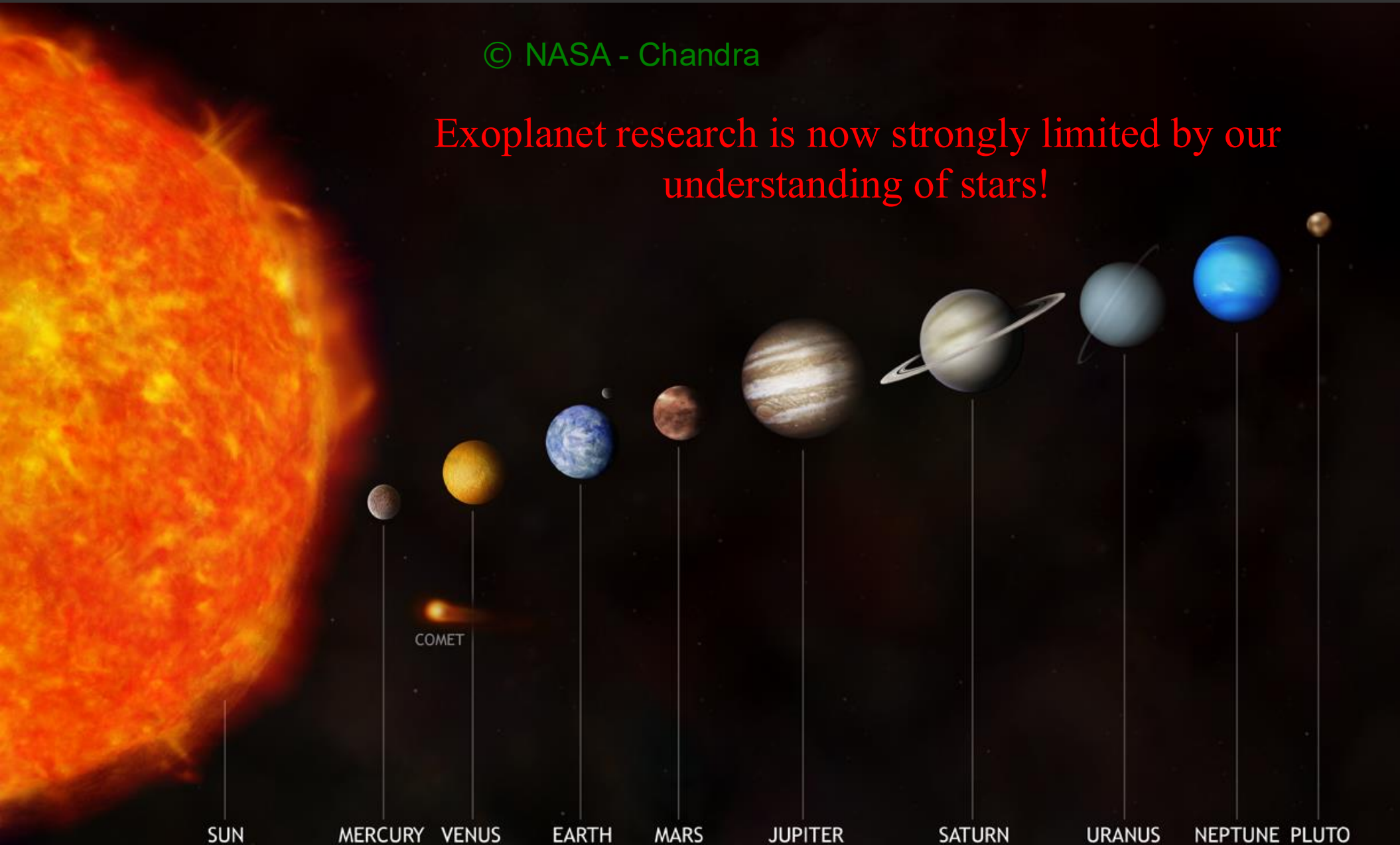
Universe age ~ 600 Myr (Tanvir et al 09, Nature: arXiv:0906.1577)

Signatures also stored in EMP stars

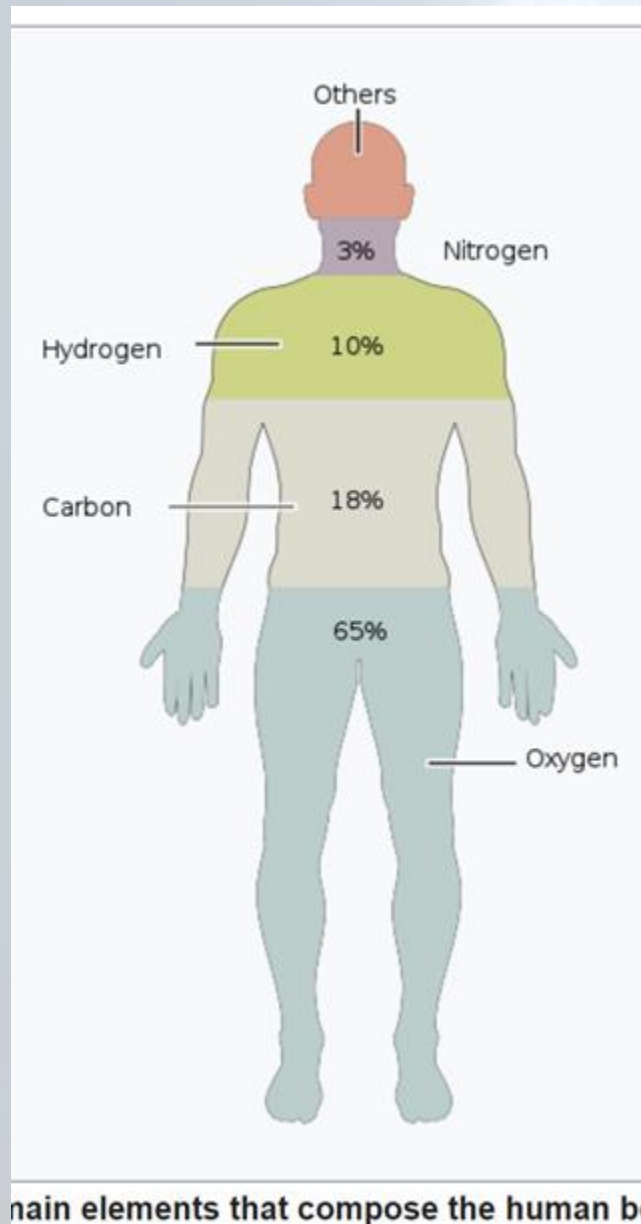
Stars as Central Object in Planetary Systems

© NASA - Chandra

Exoplanet research is now strongly limited by our understanding of stars!



What stars are you made of?



~1% traces of exotic stars

~10% Big-Bang

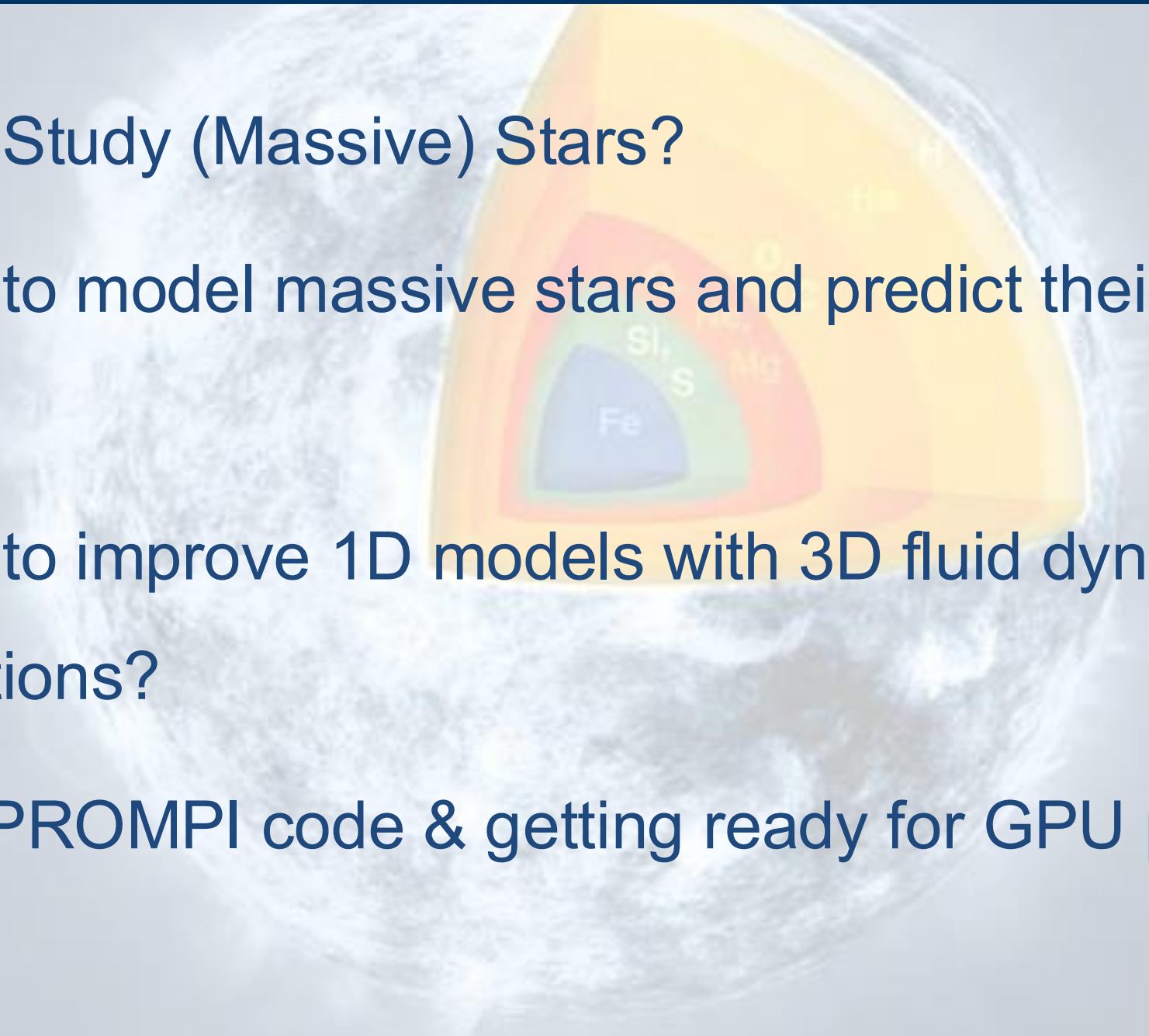
~15% low-mass stars

~74% massive stars

<https://www.youtube.com/watch?v=L8FWXNc-XxM>

Plan

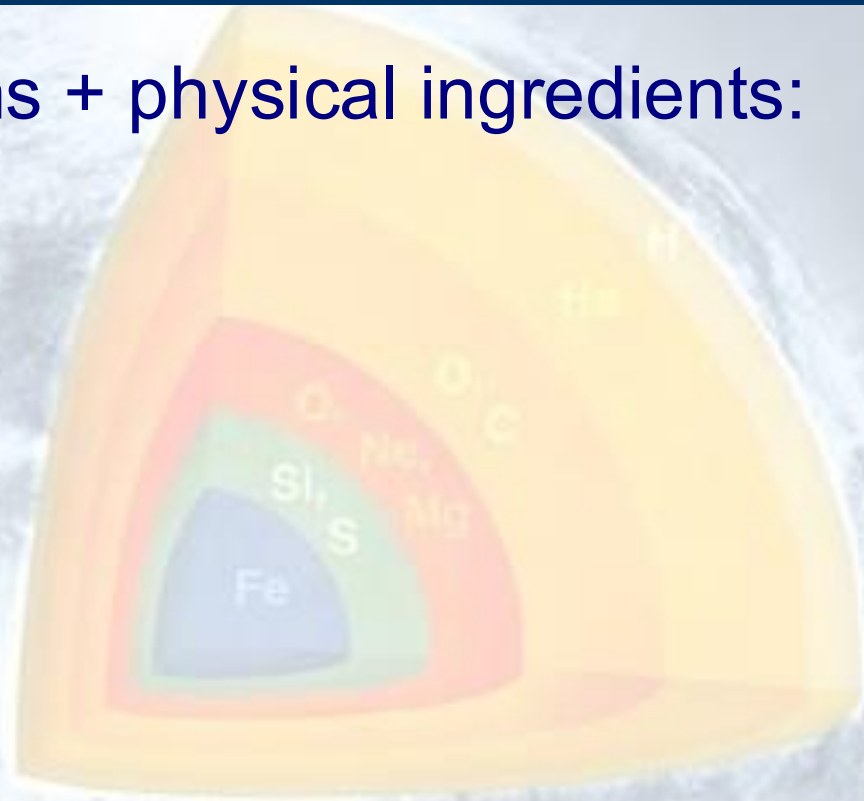
- Why Study (Massive) Stars?
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How to Model Stars?

Stellar structure equations + physical ingredients:

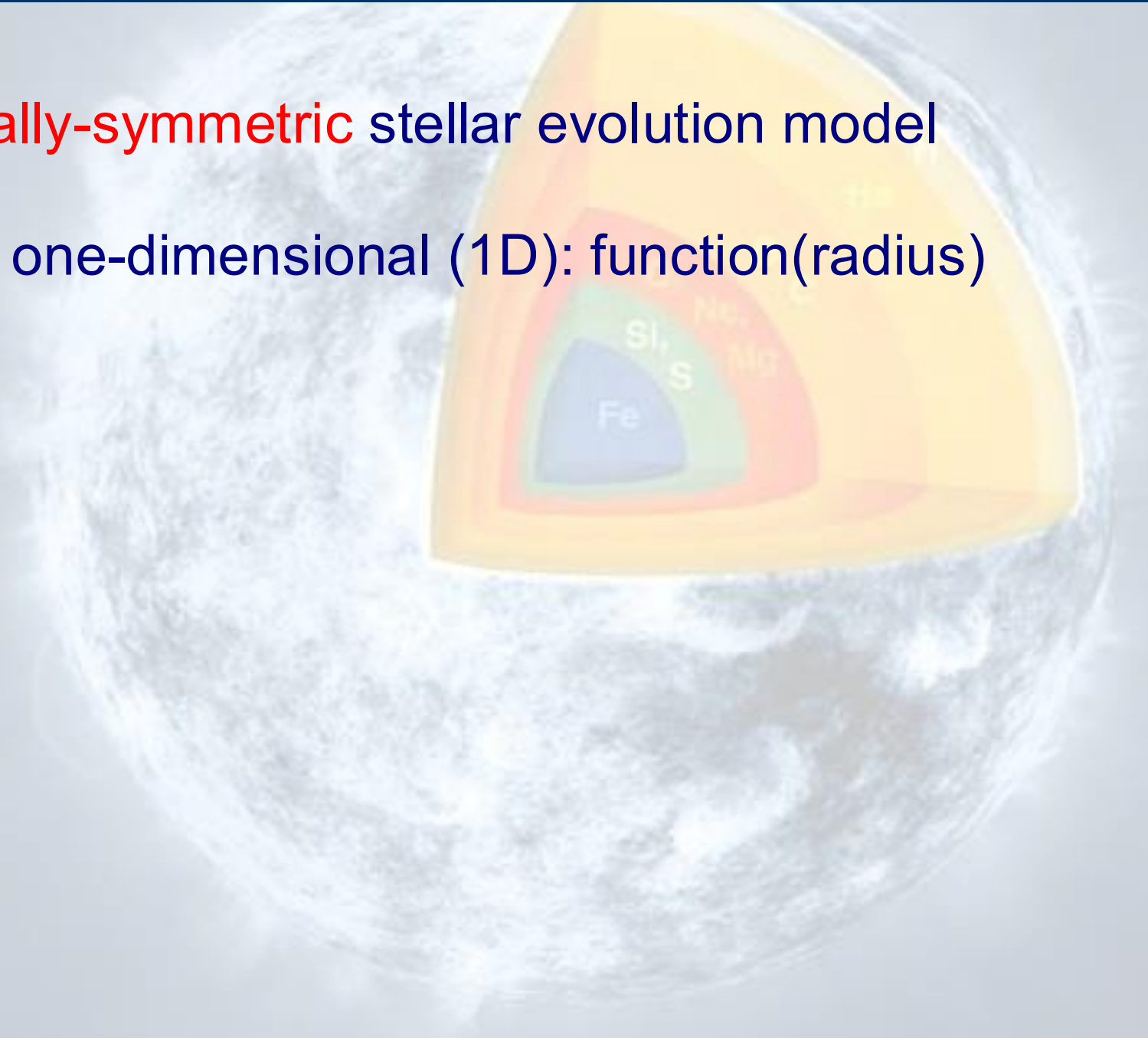
- Nuclear reactions
- Mass loss
- Convection
- Rotation
- Magnetic fields
- Binary interactions
- Equation of state, opacities & neutrino losses
including metallicity dependence



1D Stellar models: Advantages

Spherically-symmetric stellar evolution model

→ one-dimensional (1D): function(radius)



1D Stellar models: Advantages

Spherically-symmetric stellar evolution model

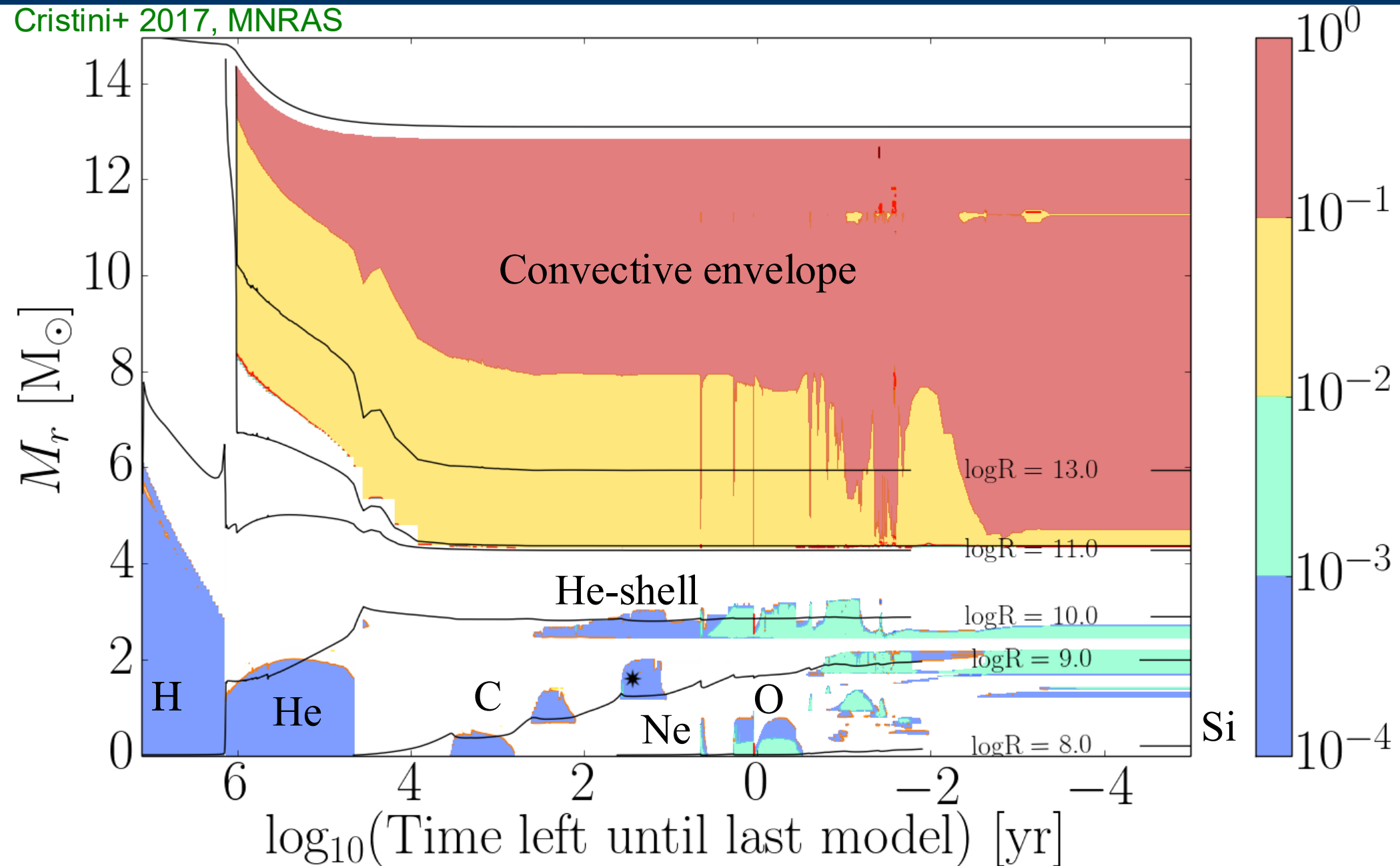
→ one-dimensional (1D): function(radius)

1D stellar evolution models have been very successful at:

- predicting the structure, evolution, fate & impact of stars
- providing a theoretical framework of interpretations of observations
- creating a bridge between astronomical observations and nuclear physics (theory and exp.)

Structure Evolution of Massive Stars

Cristini+ 2017, MNRAS



Convection takes place during most burning stages , see Jermyn+2022

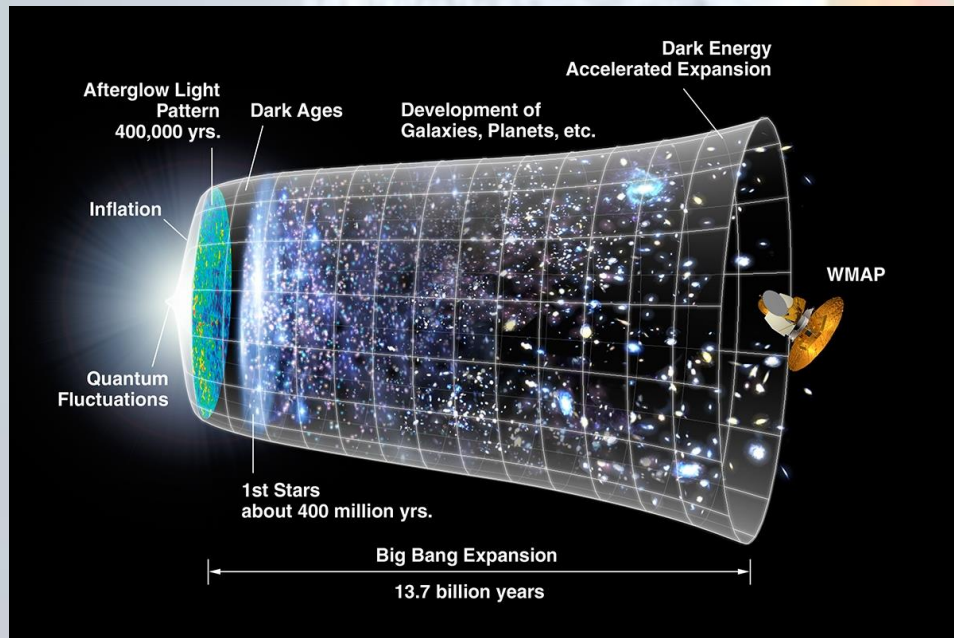
Geneva Grids of 1D Models

Large grids of models completed:

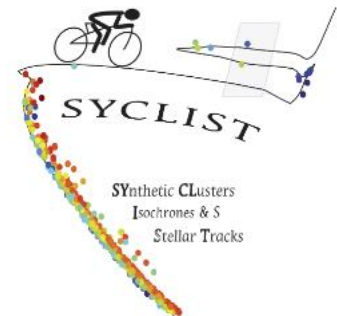
Z: 0 (first stars) $\rightarrow$ super solar

M: 0.8 or 1.7 to 120 $M_{\odot}$

With & without **rotation**

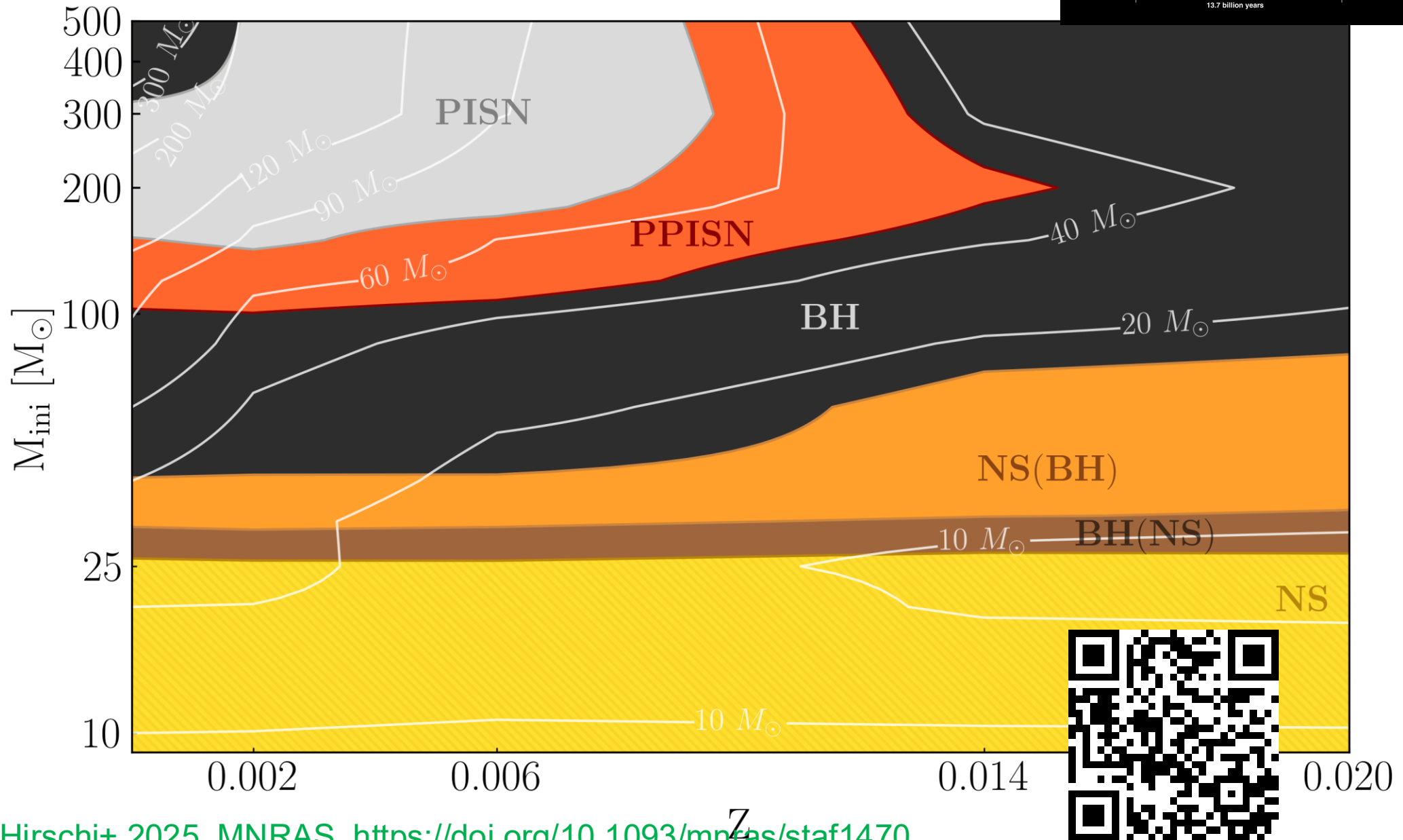
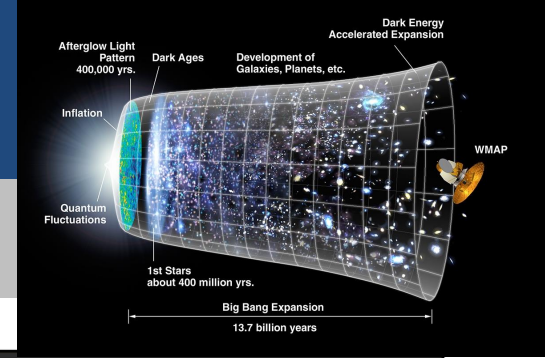


- 1 ☐ 2024A&A...690A..91S 2024/10 cited: 9
Grids of stellar models with rotation: VIII. Models from 1.7 to 500 $M_{\odot}$ at metallicity $Z = 10^{-5}$
Sibony, Yves; Shepherd, Kendall G.; Yusof, Norhasliza *and 12 more*
- 2 ☐ 2022MNRAS.511.2814Y 2022/04 cited: 36
Grids of stellar models with rotation VII: models from 0.8 to 300 $M_{\odot}$ at supersolar metallicity ($Z = 0.020$)
Yusof, Norhasliza; Hirschi, Raphael; Eggenberger, Patrick *and 11 more*
- 3 ☐ 2021A&A...652A.137E 2021/08 cited: 57
Grids of stellar models with rotation. VI. Models from 0.8 to 120 $M_{\odot}$ at a metallicity $Z = 0.006$
Eggenberger, Patrick; Ekström, Sylvia; Georgy, Cyril *and 14 more*
- 4 ☐ 2021MNRAS.501.2745M 2021/02 cited: 46
Grids of stellar models with rotation - V. Models from 1.7 to 120 $M_{\odot}$ at zero metallicity
Murphy, Laura J.; Groh, Jose H.; Ekström, Sylvia *and 10 more*
- 5 ☐ 2019A&A...627A..24G 2019/07 cited: 86
Grids of stellar models with rotation. IV. Models from 1.7 to 120 $M_{\odot}$ at a metallicity $Z = 0.0004$
Groh, J. H.; Ekström, S.; Georgy, C. *and 8 more*
- 6 ☐ 2013A&A...558A.103G 2013/10 cited: 348
Grids of stellar models with rotation. III. Models from 0.8 to 120 $M_{\odot}$ at a metallicity $Z = 0.002$
Georgy, C.; Ekström, S.; Eggenberger, P. *and 11 more*
- 7 ☐ 2012A&A...542A..29G 2012/06 cited: 262
Grids of stellar models with rotation. II. WR populations and supernovae/GRB progenitors at $Z = 0.014$
Georgy, C.; Ekström, S.; Meynet, G. *and 5 more*
- 8 ☐ 2012A&A...537A.146E 2012/01 cited: 1591
Grids of stellar models with rotation. I. Models from 0.8 to 120 $M_{\odot}$ at solar metallicity ($Z = 0.014$)
Ekström, S.; Georgy, C.; Eggenberger, P. *and 9 more*



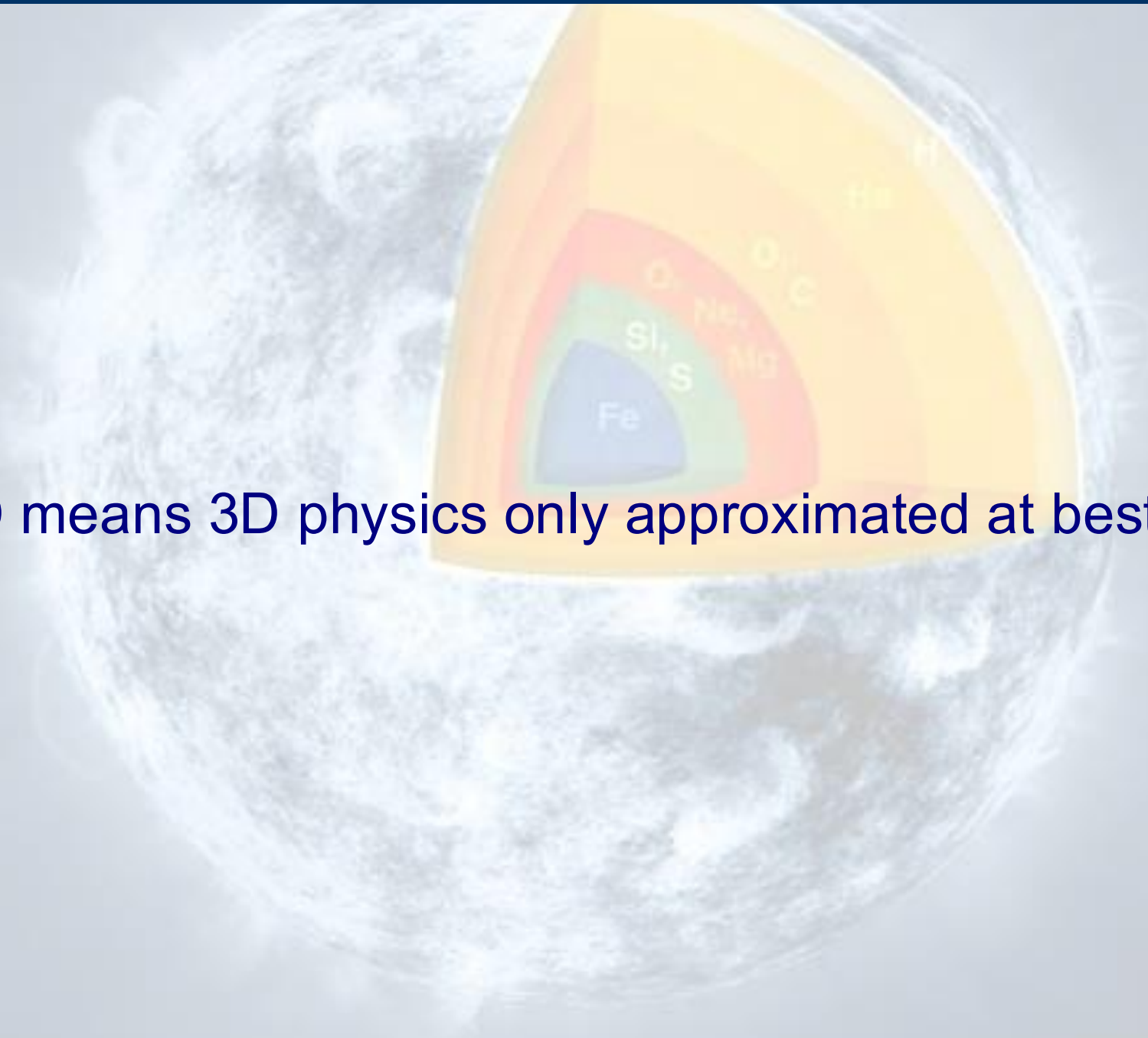
Fate($M, Z, \Omega=0$)

Mass loss and final masses strongly depend on metallicity (Z)

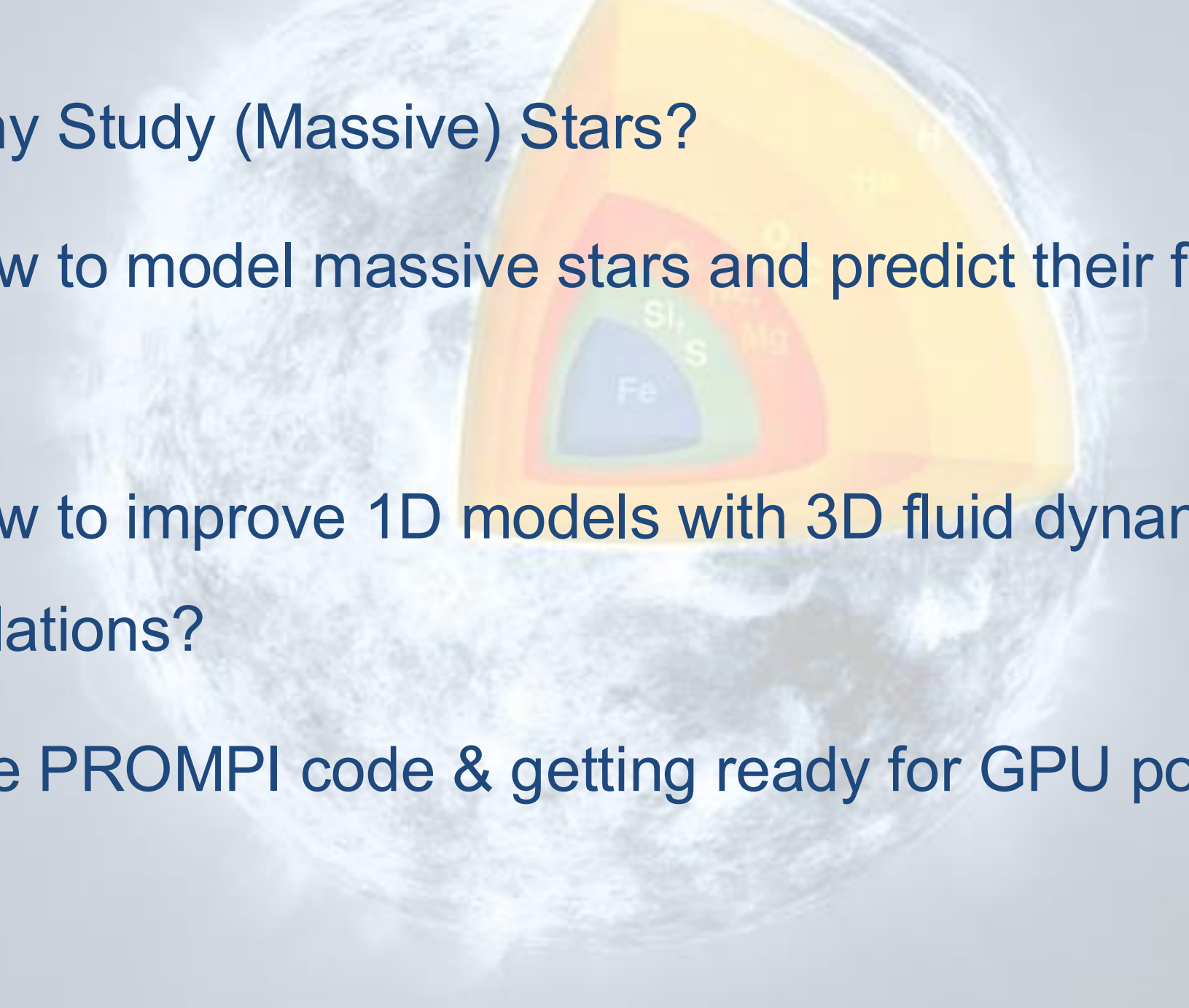


1D Stellar models: Challenges

1D means 3D physics only approximated at best



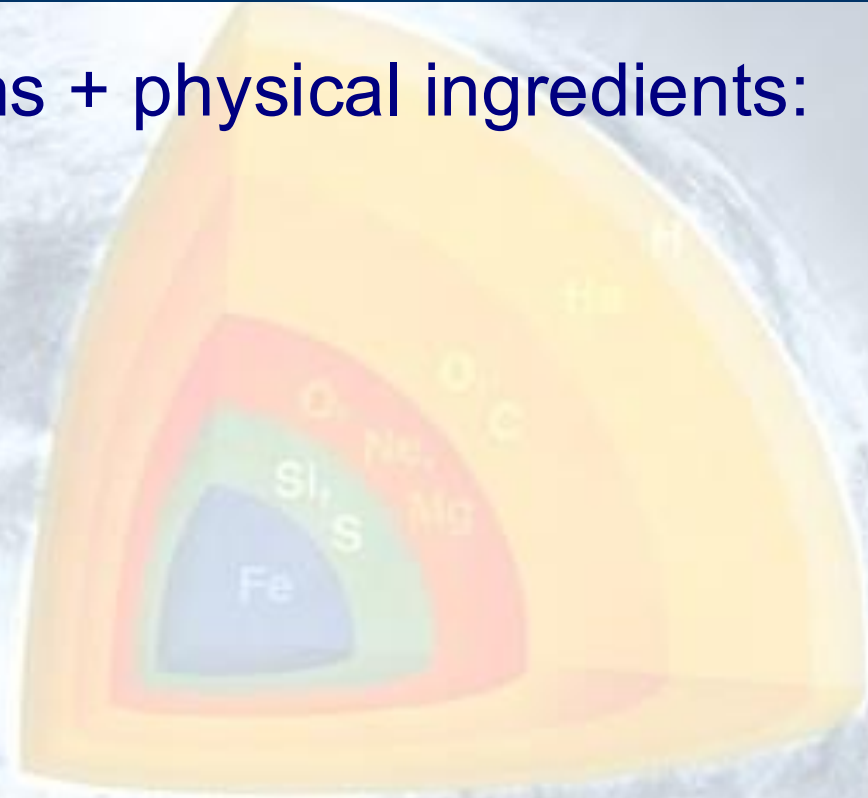
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How to Model Stars in 3D?

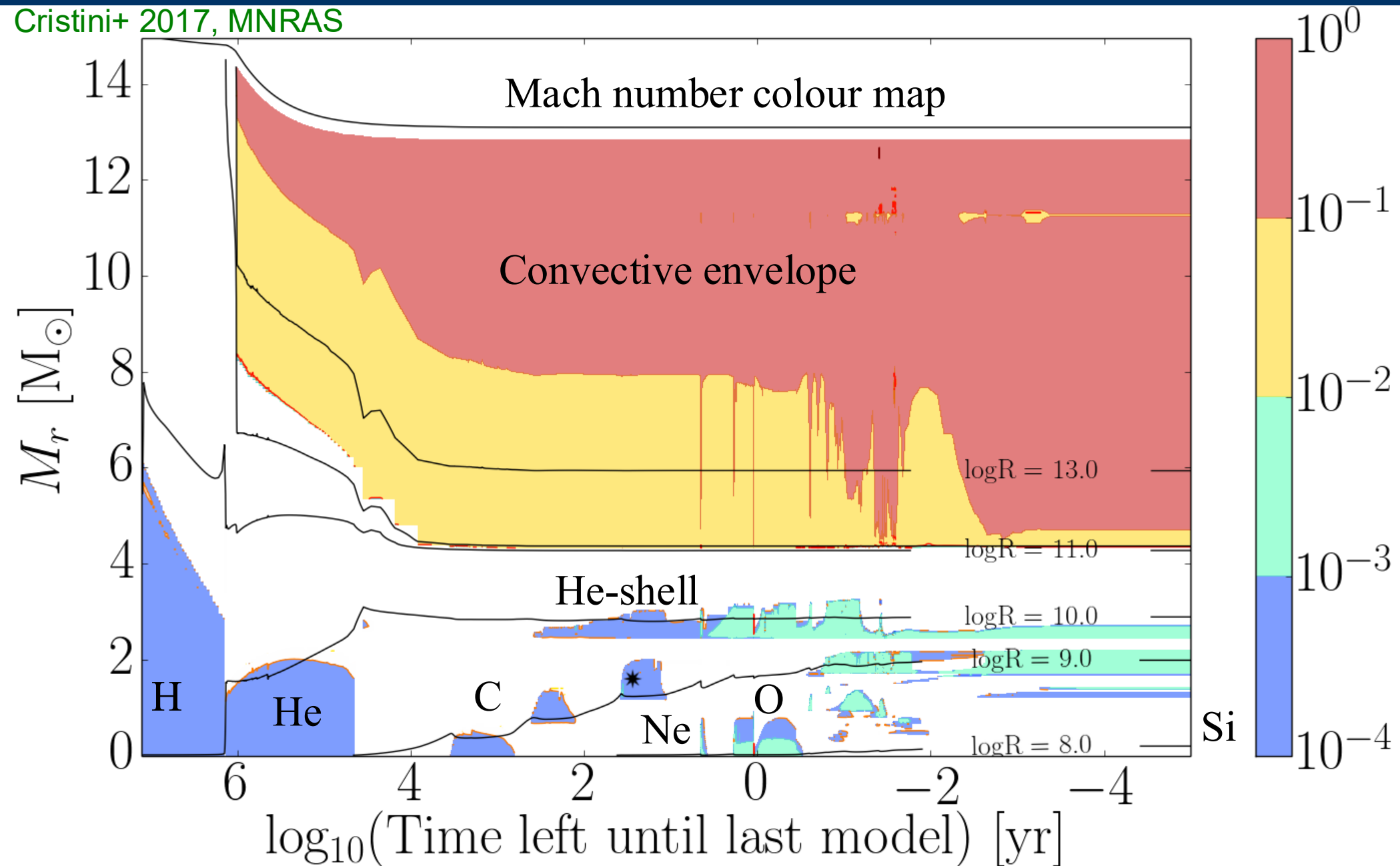
Stellar structure equations + physical ingredients:

- Nuclear reactions
- Mass loss
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Carbon-burning 32(to)1D loop

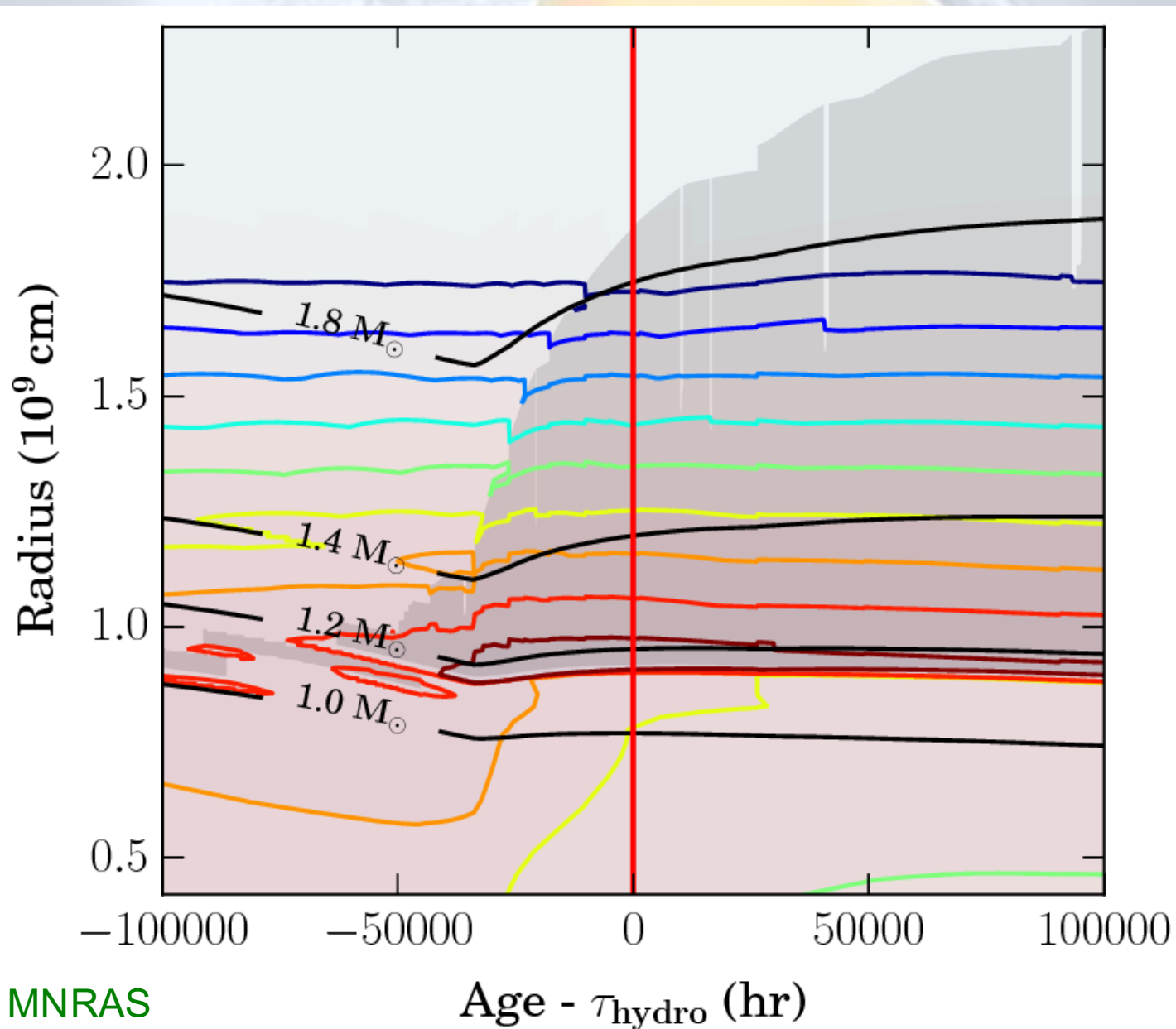
Cristini+ 2017, MNRAS



Convection takes place during most burning stages

From 1D to 3D

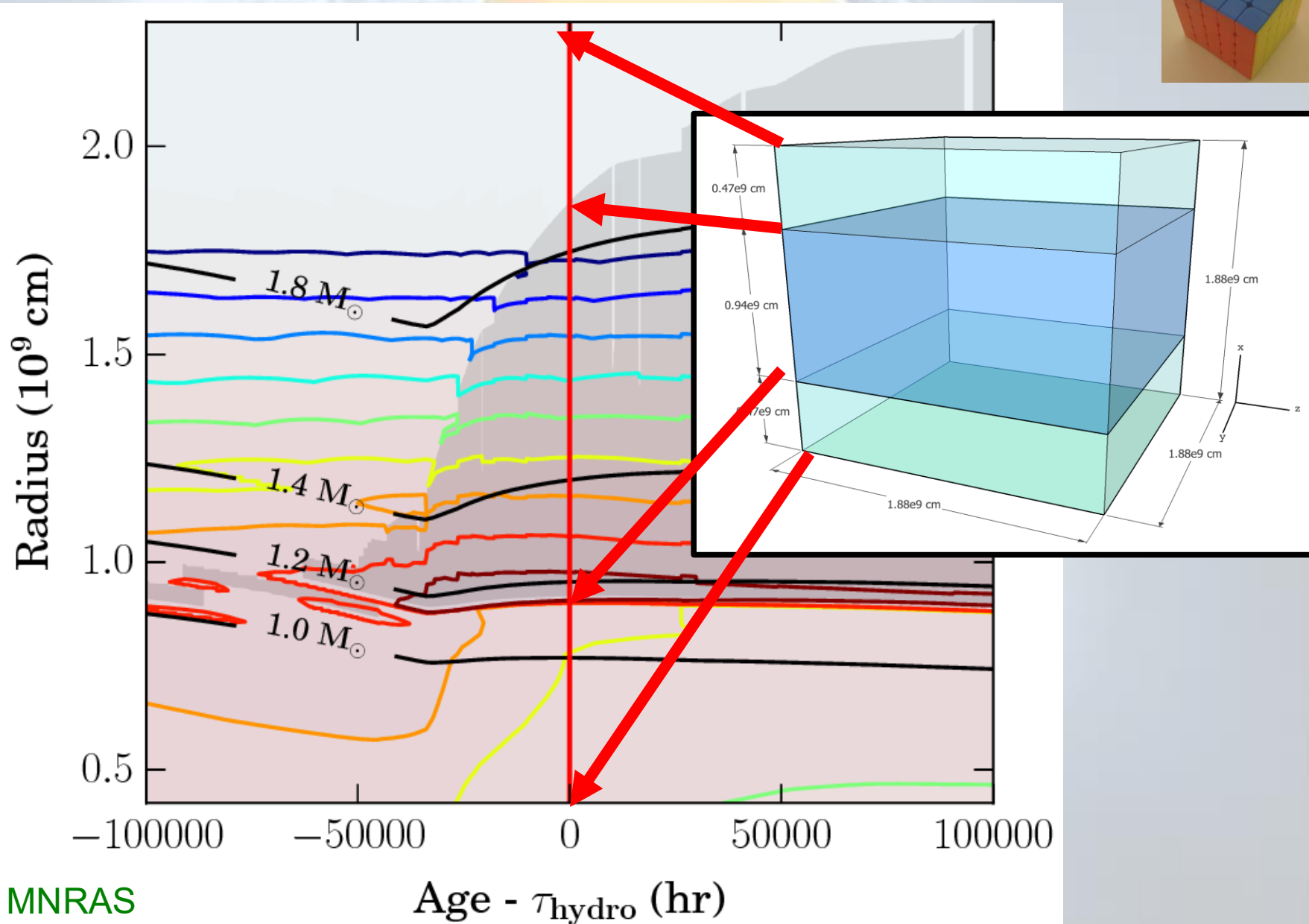
C-shell in $15 M_{\odot}$, $Z=0.014$ 1D stellar evolution model



From 1D to 3D

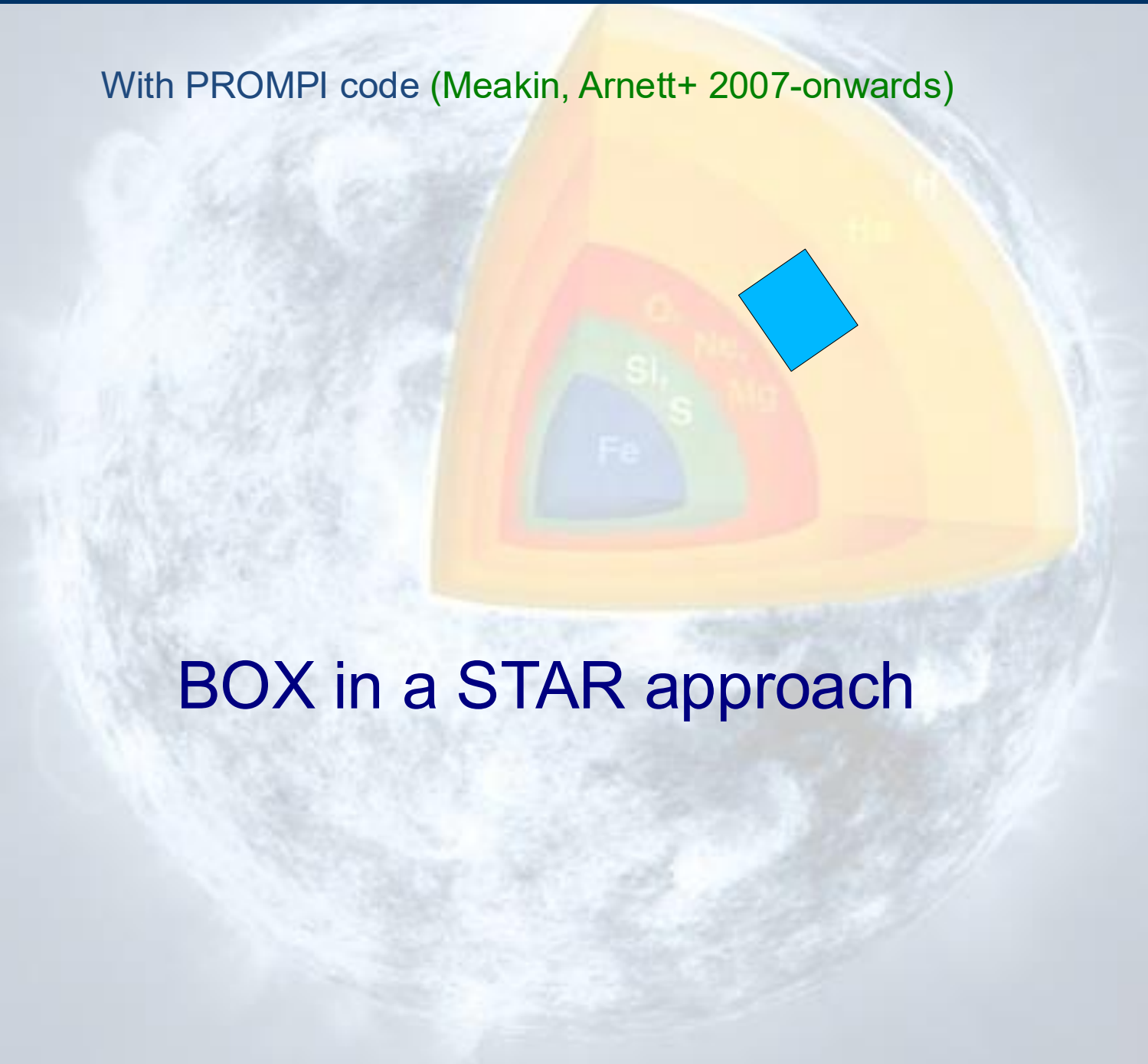
<https://www.youtube.com/@ezrahirsch/>

C-shell in $15 M_{\odot}$, $Z=0.014$ 1D stellar evolution model



3D simulations of Carbon burning

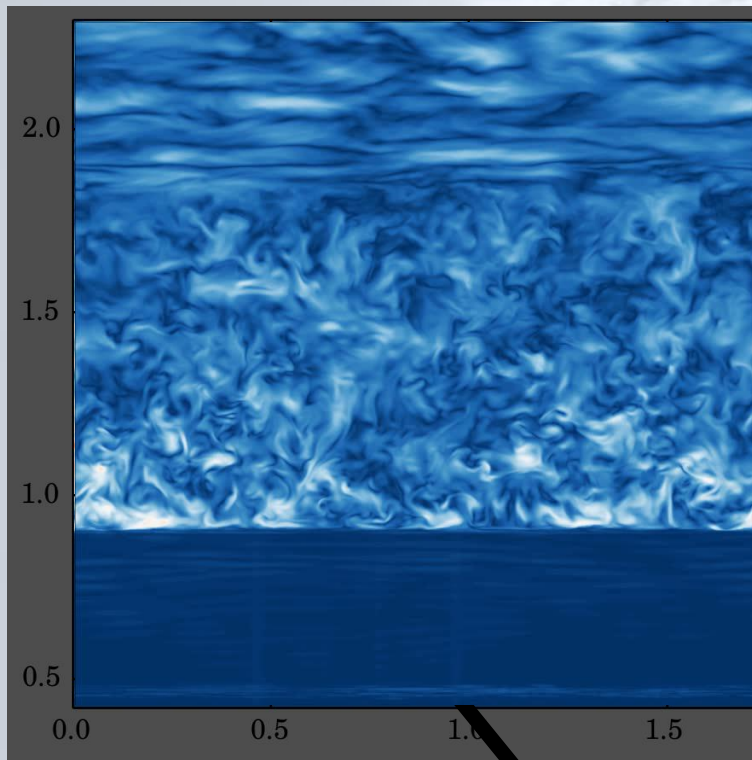
With PROMPI code (Meakin, Arnett+ 2007-onwards)



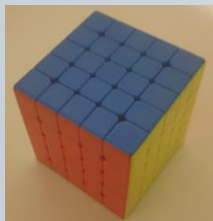
BOX in a STAR approach

1 to 3 to 1D: first 321D loop

Targeted 3D simulations

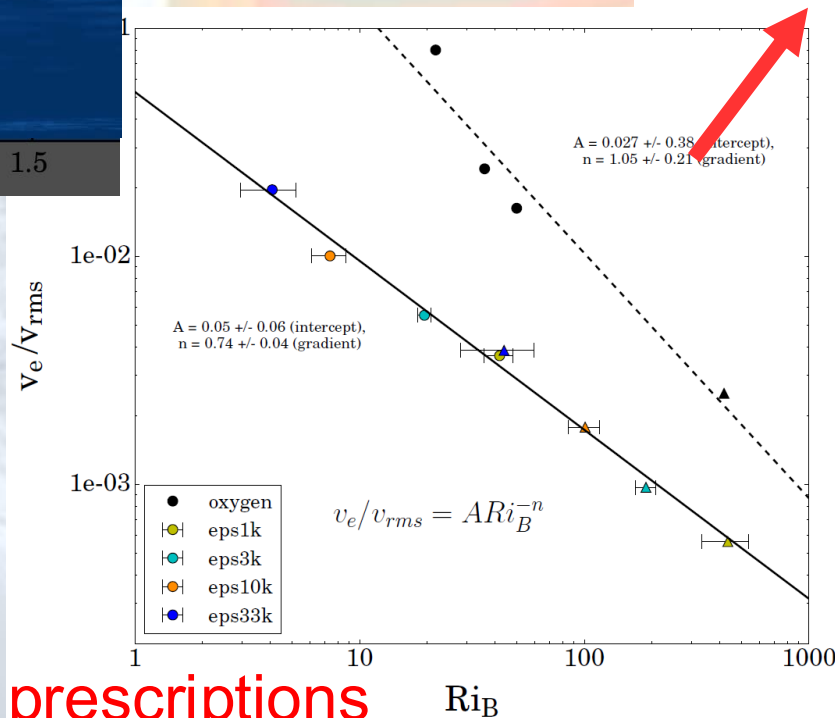
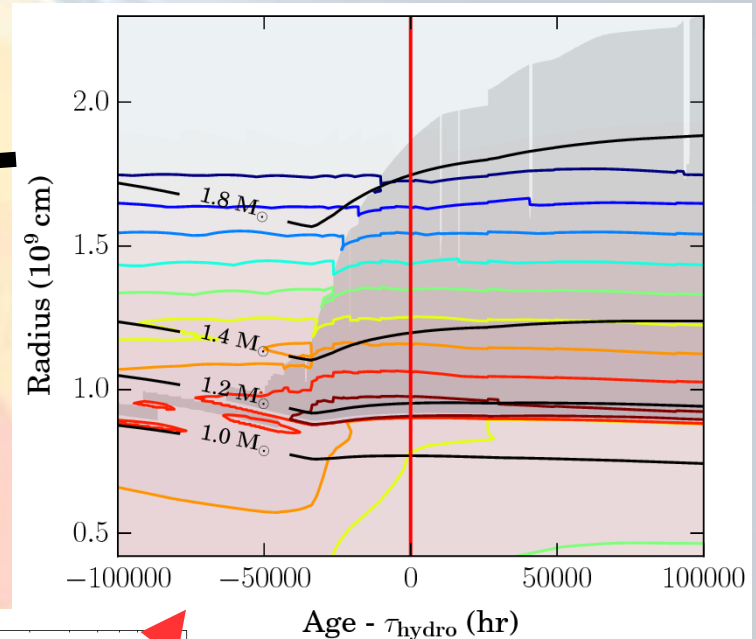


Cristini+2017



<https://www.youtube.com/@ezrahirschil/>

Uncertainties in 1D

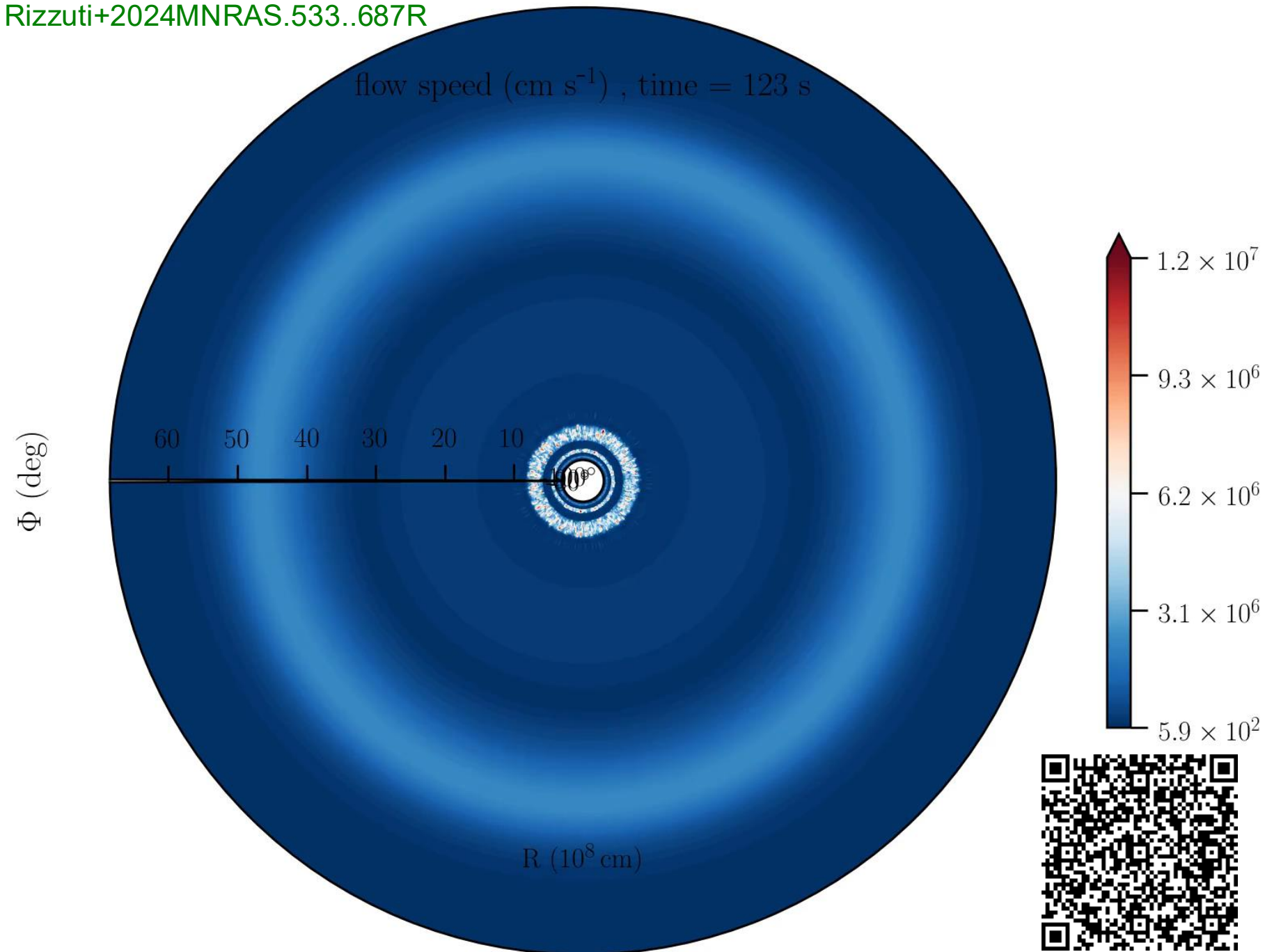


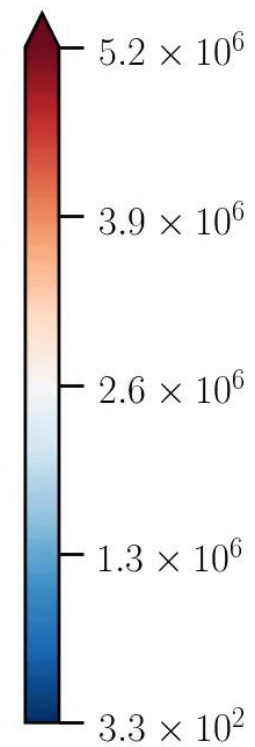
Cristini+2019 (C)
 Rizzuti+2022 (Ne)

→ Improve theoretical prescriptions

Towards 4-pi & shell mergers

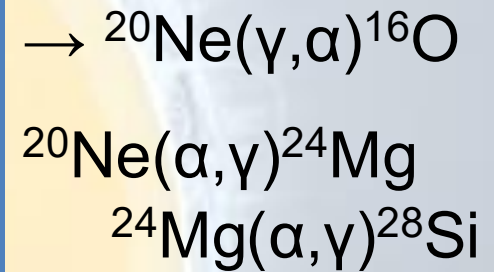
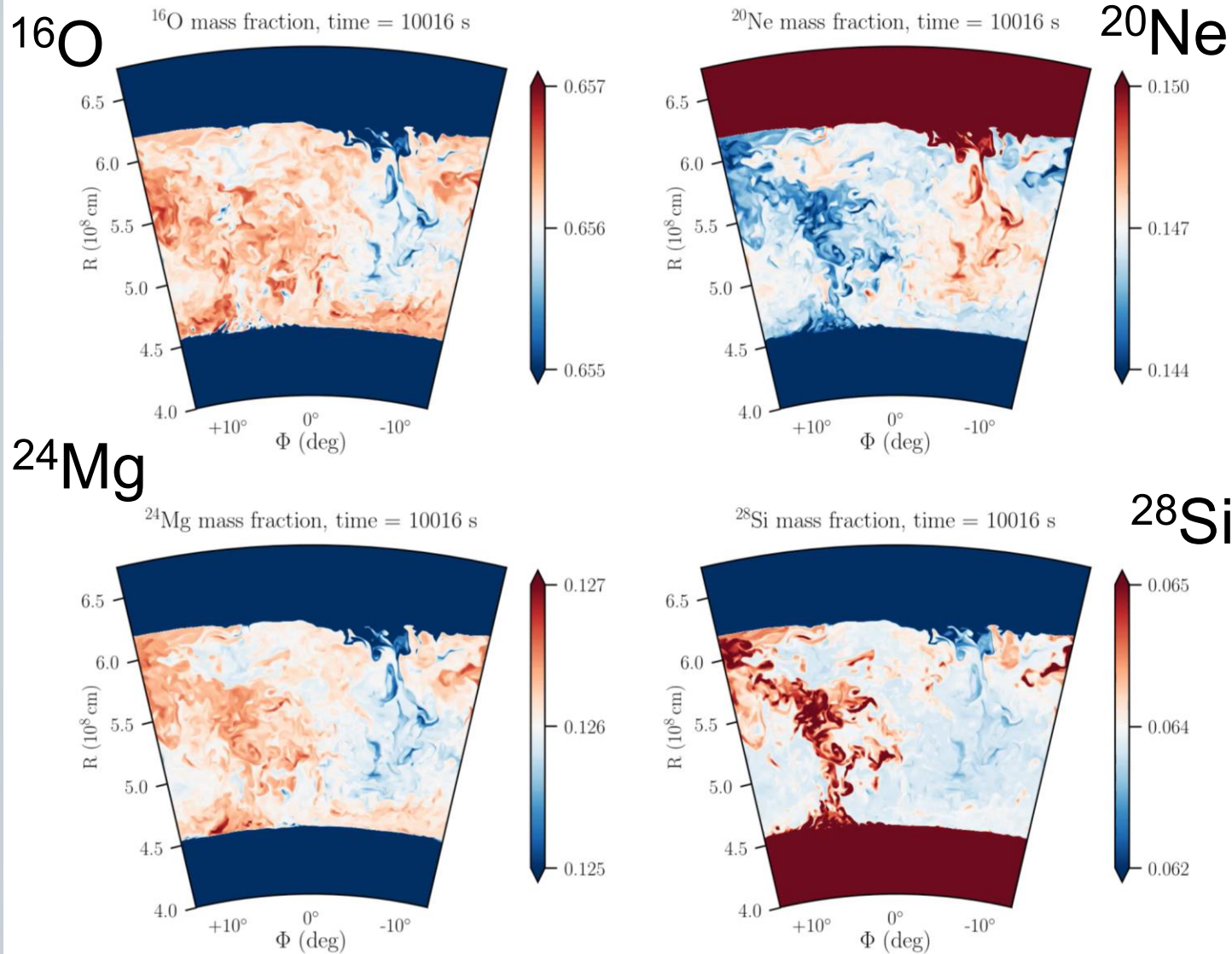
Rizzuti+2024MNRAS.533..687R



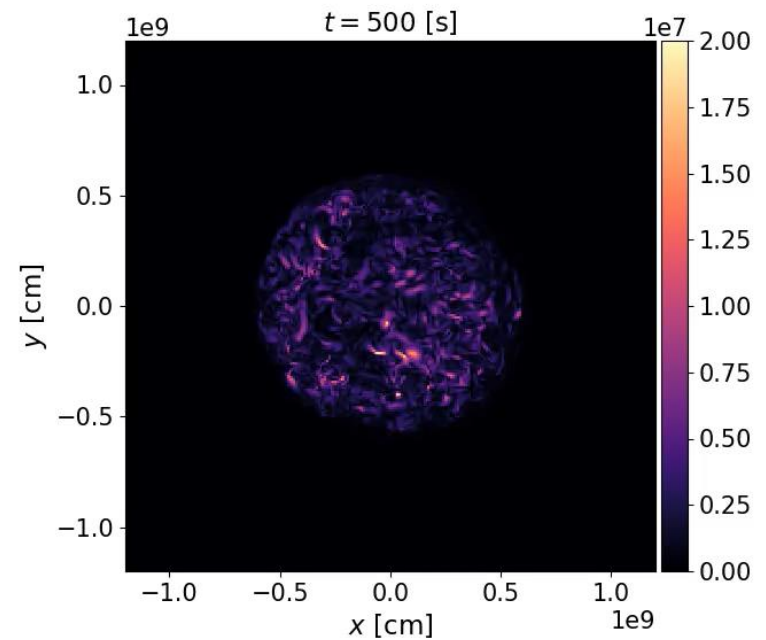
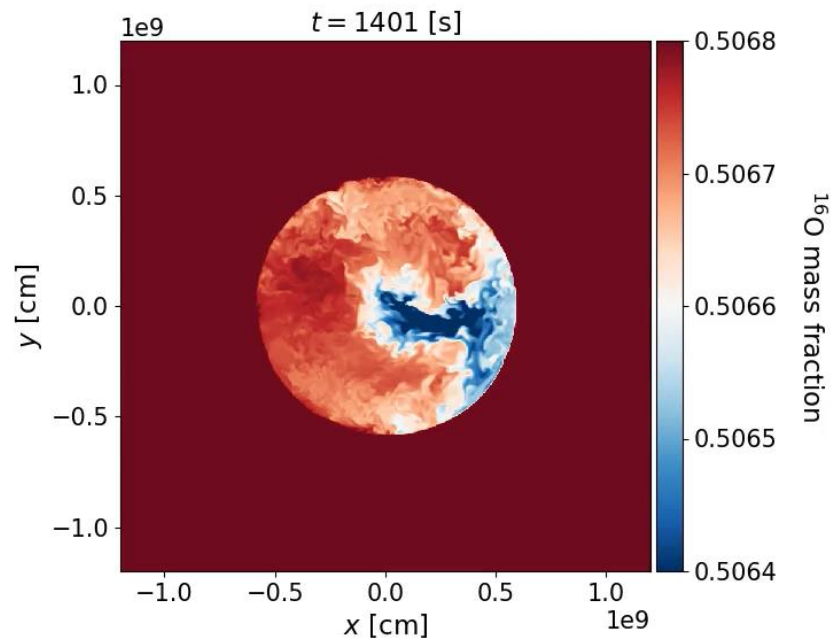


3D Nuclear Astrophysics

Rizzuti, RH, etal, 2023MNRAS.523.2317R



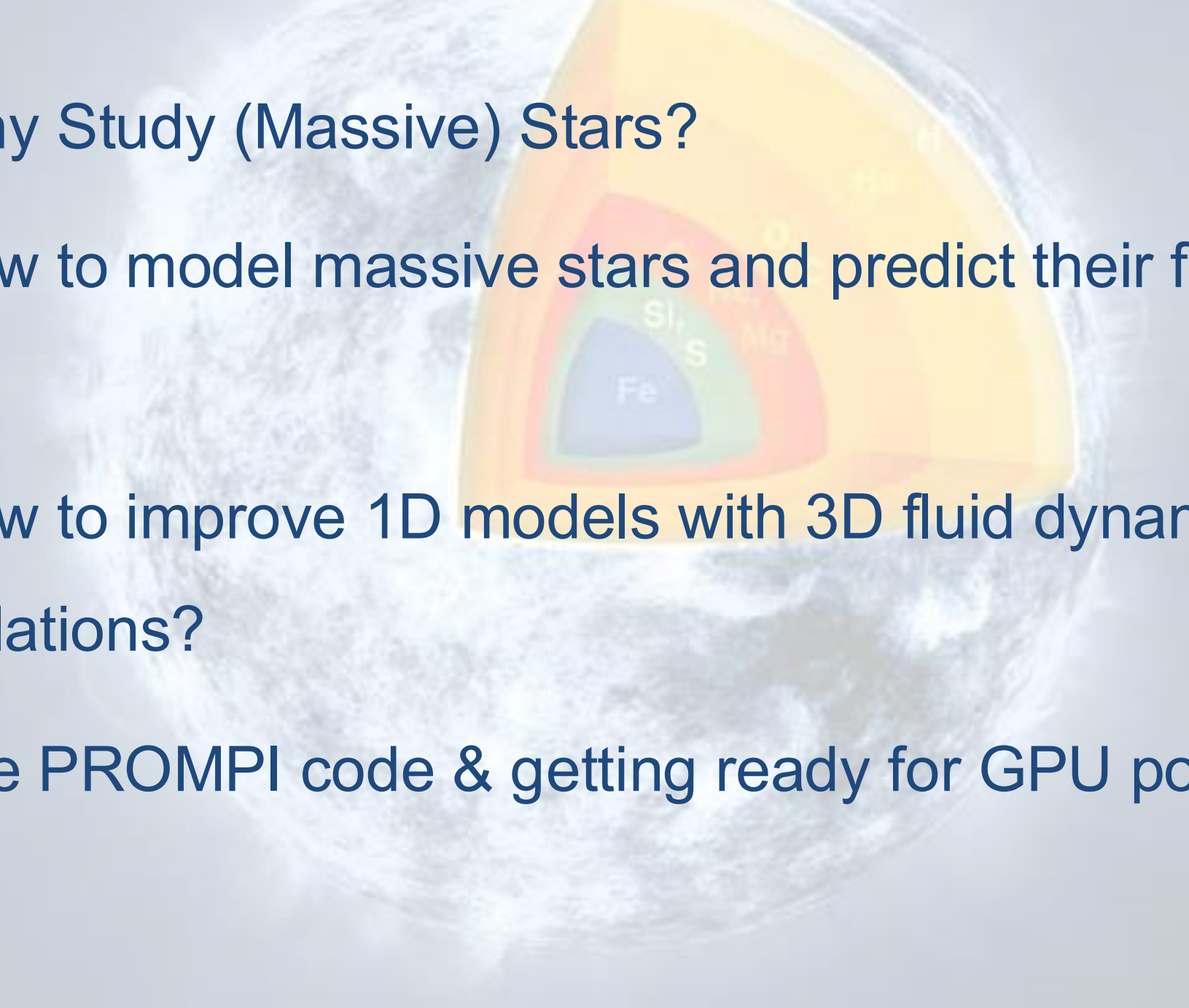
3D Nuclear Astrophysics & B-fields



Leidi+, Rizzuti+ in prep.



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PROMPI Code

- Brief overview
- Time integration & spatial discretisation
- Parallelism & domain decomposition
- Getting ready for GPU porting

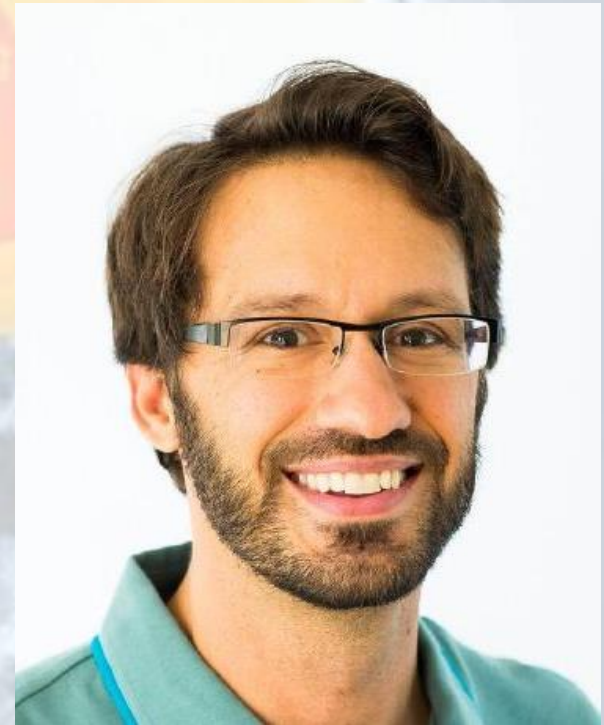


PROMPI Code

Dave Arnett



Casey Meakin



PROMPI 3D Hydrodynamics ILES Code

Reactive Euler Equations: PROMPI

$$\frac{\partial \mathbf{Q}}{\partial t} + \nabla \cdot \mathbf{\Phi} = \mathbf{S}, \quad (1)$$

with the *state vector*

$$\mathbf{Q} \equiv \begin{bmatrix} \rho \\ \rho \mathbf{u} \\ \rho E \\ \rho X_i \end{bmatrix}, \quad (2)$$

the *flux vector*

$$\mathbf{\Phi} \equiv \begin{bmatrix} \rho \mathbf{u} \\ \rho \mathbf{u} \mathbf{u} + p \\ (\rho E + p) \mathbf{u} + \mathbf{F}_r \\ \rho X_i \mathbf{u} \end{bmatrix}, \quad (3)$$

and the *source vector*

$$\mathbf{S} \equiv \begin{bmatrix} 0 \\ \rho \mathbf{g} \\ \rho \mathbf{u} \cdot \mathbf{g} + \rho \epsilon_{net} \\ R_i \end{bmatrix}. \quad (4)$$

where $E = e_{int} + e_{ke}$ is the total energy per gram and ρ , p , $\mathbf{u}$, $\mathbf{g}$, and T are the density, pressure, velocity, gravitational force field and temperature.

Core Solver

- Fully compressible PPM Hydrodynamics, multi-species advection based on PROMETHEUS
- Cartesian, Spherical, Cylindrical Coordinates
- Domain decomposition for Parallel Computing (Message Passing with MPI)

Added Physics (Operator Splitting)

- Shellular Gravity
- Helmholtz Equation of State (Timmes & Swesty, 2000)
- General Nuclear Reaction Network
- Plasma Neutrino Cooling Rates (Itoh et al.)
- Radiative Transfer: Diffusive, OPAL Opacities (Implicit/Explicit Solver)

Initial Value Problem

- Initialization Routines: HSE Integrator, Star Mapper

PROMPI : Meakin, Arnett+ 2007-onwards

PROMETHEUS : Fryxell, Mueller, Arnett 1989

PPM method : Colella & Woodward 1984

Multi-D Stellar Evolution, Casey A. Meakin, July 2006

Time Integration

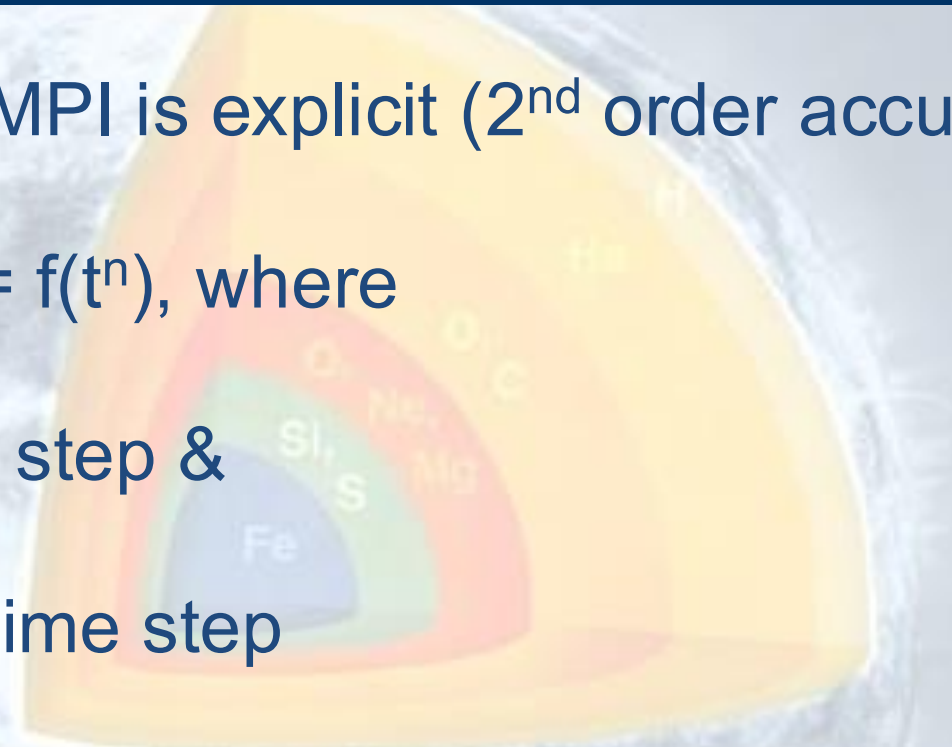
- Time integration PROMPI is explicit (2nd order accurate):

$$dx/dt = [x(t^{n+1}) - x(t^n)] / \Delta t = f(t^n), \text{ where}$$

t^n : old/known time step &

$t^{n+1} = t^n + \Delta t$: new time step

- Key advantage: easier to program & parallelise compared to time implicit method ($f(t^{n+1})$)
- Main limitation: time step limited by the sound speed (CFL condition) so not ideal for very low-Mach N° flows ($Ma < 10^{-3}$)



Spatial Discretisation

- PROMPI uses PPM method (Colella & Woodward 1984)
- Generally, new quantities (t^{n+1}) at cell indices $x(i,j,k,t^{n+1})$ depend only on known quantities (t^n) at nearby cells, e.g. (i,j,k,t^n) , $(i+1,j,k,t^n)$, $(i,j-2,k,t^n)$ or $(i,j,k-1,t^n)$:

Fryxell et al 2000 (FLASH code)

$$\langle \rho \rangle_i^{n+1} = \langle \rho \rangle_i^n - \frac{\Delta t}{\Delta V_i} (A_{i+1/2} \bar{\rho}_{i+1/2} \bar{u}_{i+1/2} - A_{i-1/2} \bar{\rho}_{i-1/2} \bar{u}_{i-1/2}), \quad (92)$$

$$\begin{aligned} \langle \rho \rangle_i^{n+1} \langle u \rangle_i^{n+1} &= \langle \rho \rangle_i^n \langle u \rangle_i^n - \frac{\Delta t}{\Delta V_i} (A_{i+1/2} \bar{\rho}_{i+1/2} \bar{u}_{i+1/2}^2 - A_{i-1/2} \bar{\rho}_{i-1/2} \bar{u}_{i-1/2}^2) - \frac{\Delta t}{\Delta \xi} (\bar{P}_{i+1/2} - \bar{P}_{i-1/2}) \\ &\quad + \frac{\Delta t}{2} (\langle \rho \rangle_i^n \langle g \rangle_i^n + \langle \rho \rangle_i^{n+1} \langle g \rangle_i^{n+1}), \end{aligned} \quad (93)$$

$$\langle \rho \rangle_i^{n+1} \langle v \rangle_i^{n+1} = \langle \rho \rangle_i^n \langle v \rangle_i^n - \frac{\Delta t}{\Delta V_i} (A_{i+1/2} \bar{\rho}_{i+1/2} \bar{v}_{i+1/2} \bar{u}_{i+1/2} - A_{i-1/2} \bar{\rho}_{i-1/2} \bar{v}_{i-1/2} \bar{u}_{i-1/2}), \quad (94)$$

$$\langle \rho \rangle_i^{n+1} \langle w \rangle_i^{n+1} = \langle \rho \rangle_i^n \langle w \rangle_i^n - \frac{\Delta t}{\Delta V_i} (A_{i+1/2} \bar{\rho}_{i+1/2} \bar{w}_{i+1/2} \bar{u}_{i+1/2} - A_{i-1/2} \bar{\rho}_{i-1/2} \bar{w}_{i-1/2} \bar{u}_{i-1/2}), \quad (95)$$

$$\begin{aligned} \langle \rho \rangle_i^{n+1} \langle E \rangle_i^{n+1} &= \langle \rho \rangle_i^n \langle E \rangle_i^n - \frac{\Delta t}{\Delta V_i} [A_{i+1/2} (\bar{\rho}_{i+1/2} \bar{E}_{i+1/2} + \bar{P}_{i+1/2}) \bar{u}_{i+1/2} - A_{i-1/2} (\bar{\rho}_{i-1/2} \bar{E}_{i-1/2} + \bar{P}_{i-1/2}) \bar{u}_{i-1/2}] \\ &\quad + \frac{\Delta t}{2} (\langle \rho \rangle_i^n \langle u \rangle_i^n \langle g \rangle_i^n + \langle \rho \rangle_i^{n+1} \langle u \rangle_i^{n+1} \langle g \rangle_i^{n+1}), \end{aligned} \quad (96)$$

$$\langle \rho \rangle_i^{n+1} \langle X_\ell \rangle_i^{n+1} = \langle \rho \rangle_i^n \langle X_\ell \rangle_i^n - \frac{\Delta t}{\Delta V_i} (A_{i+1/2} \bar{\rho}_{i+1/2} \bar{X}_{\ell i+1/2} \bar{u}_{i+1/2} - A_{i-1/2} \bar{\rho}_{i-1/2} \bar{X}_{\ell i-1/2} \bar{u}_{i-1/2}). \quad (97)$$

Note: 1/2
indices
refer to
cell
interfaces

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https://gitlab.com/prompi/prompi/-/blob/develop/simulator/HYDRO/hydrow/hydrow_update_field_variables.f90

```
29      !$acc parallel loop collapse(2) independent private(ugmax)
30      do k = 1, nk
31          do j = 1, nj
32              ugmax = 0.d00
33              do i = 1, nzn8
34                  ugmax = dmax1(0.d00, dabs(ugrid(i, j, k)), dabs(ugrdl(i, j, k)))
35              enddo
36
37              do i = 5, nzn4
38                  scrch1(i, j, k) = dvol(i, j, k) / dvol(i, j, k)
39                  scrch1(i, j, k) = cvmgz(1.d00, scrch1(i, j, k), ugmax)
40                  dtdx(i, j, k) = dt / dvol(i, j, k)
41                  rhonu(i, j, k) = rho(i, j, k) * scrch1(i, j, k) - dtdx(i, j, k) * (rhoFlx(i + 1, j, k) - rhoFlx(i, j, k))
42                  rhonu(i, j, k) = dmax1 (smlrho, rhonu(i, j, k))
43              enddo
44          end do
45      end do
```

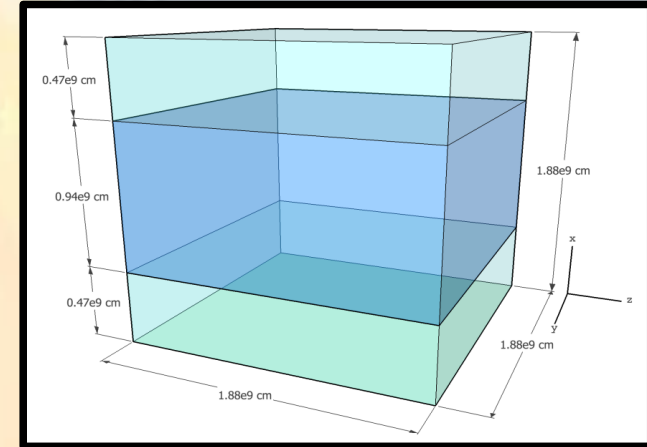
So work for different cells can “easily” be distributed across different CPUs

PROMPI Parallelism (MPI) & Domain Decomposition

Example for cartesian coordinates and cubic domain:

Entire domain: n^3 cells

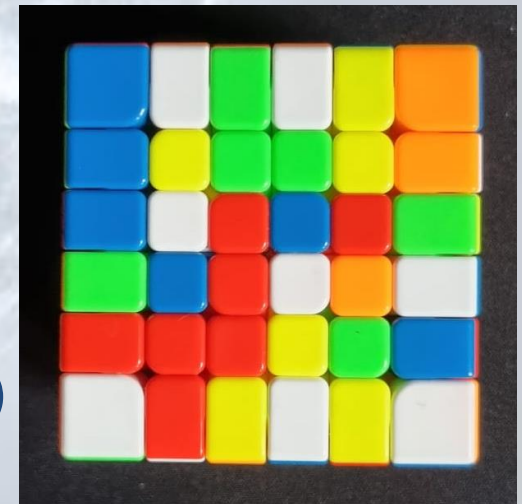
Divided into n_{tiles}^3 MPI sub-domains
of size $q^3 = (n/n_{\text{tiles}})^3$ cells



($n = 2, n_{\text{tiles}} = 2 \rightarrow q=1$)



($n = 6, n_{\text{tiles}} = 6 \rightarrow q=1$)



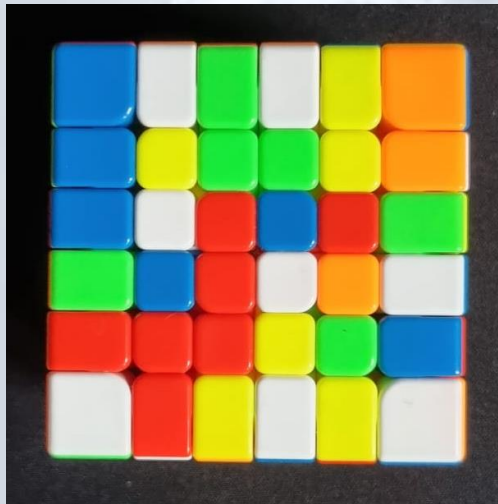
Goals:

- Larger volume
- Increase n (better resolution)
- Increase n_{tiles} (faster calculation)
- Stay efficient

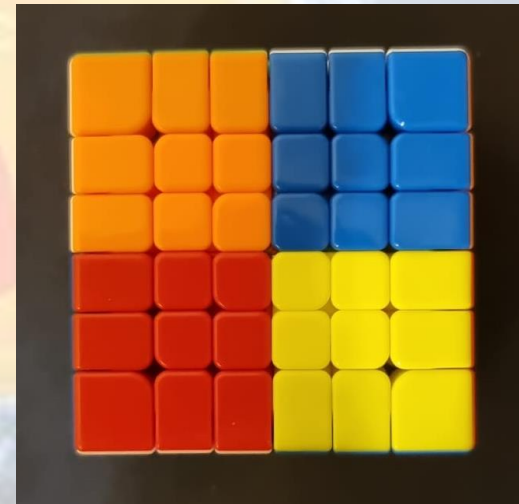
PROMPI Parallelism (MPI) & Domain Decomposition

But information at interfaces needs to be shared (MPI) between sub-domains so q cannot be too small (n° of ghost cells $\uparrow$ with n_{tiles}).

($n = 6, n_{\text{tiles}} = 6 \rightarrow q=1$)



($n = 6, n_{\text{tiles}} = 2 \rightarrow q=3$)



Which is better?

OR

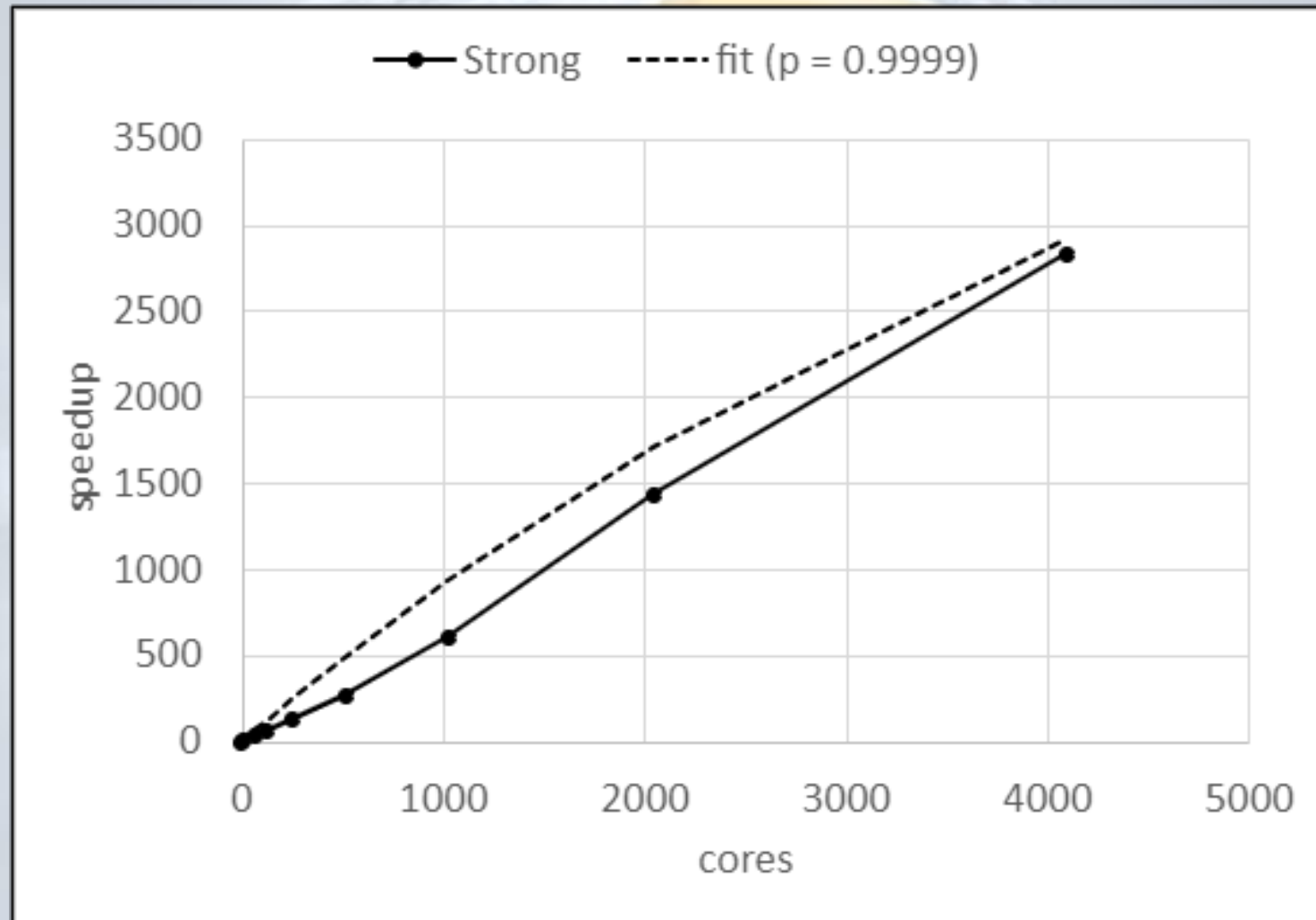
At the same time, larger q means larger RAM footprint so q cannot be too large.

Different limitations/constraints considerations:

- communications overheads ($q > 32$ in PROMPI on CPU core)
- RAM limits ($q < \sim 1024$ on CPU cores)

Profiling is critical

PROMPI Code: Strong Scaling on Cosma8



speed-up, $s = 1/[(1-p)+p/N]$, where p is the fraction of the code that is parallel and thus scales perfectly and N is the number of cores used in the calculation relative to 64 CPU cores:

For 4096 cores, ideal speed-up is 4096, actual one is around 2800, or 68% of ideal

Can we go further/faster with GPUs?

MPI + OpenACC (Fortran) on NVIDIA GPUs

MPI + OpenMP (Fortran) on AMD MI300A is an interesting alternative

$$(n = 6, n_{\text{tiles}} = 2 \rightarrow q=3)$$

MPI sub-domains
(colour) &

OpenMP/
OpenACC
Sub-sub-domains
(pieces)

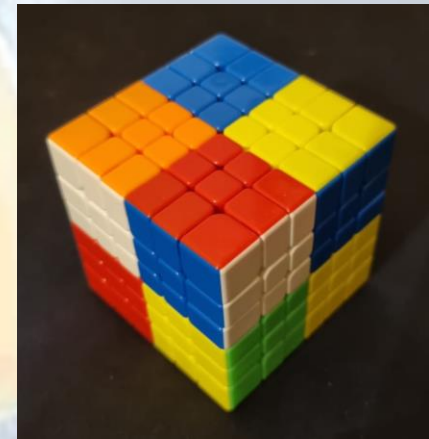
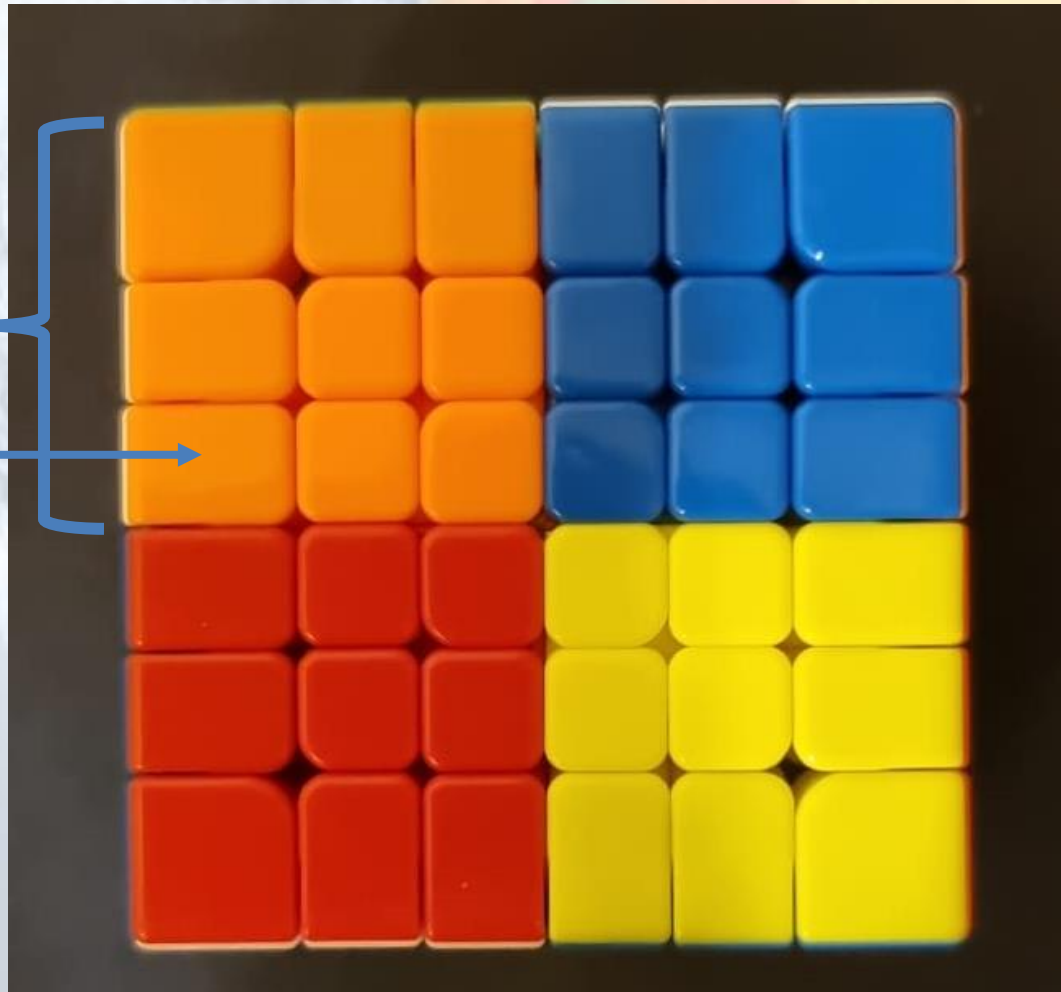


IMAGE CREDITS:
[@ezrahirschi](#)
([YouTube.com](#))

Can we go further/faster with GPUs?

MPI + Expectations:

- MPI +
- GPU has more computing power than a CPU core: better if q is higher
 - GPU generally has smaller memory than a CPU core: so q cannot be too high

Narrower range of efficient q values: 200-300

Not trivial to use GPU efficiently (2 levels of parallelism needed)

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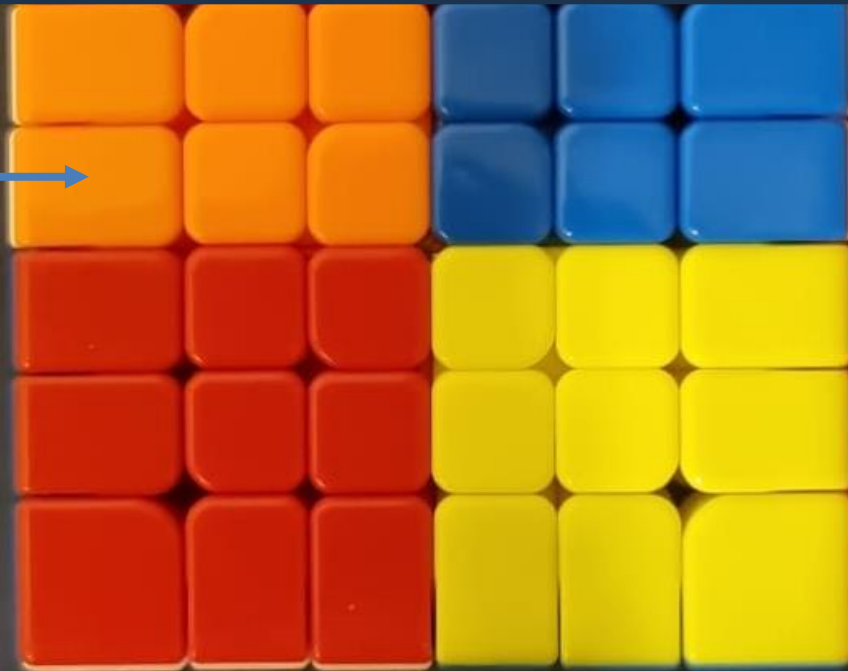


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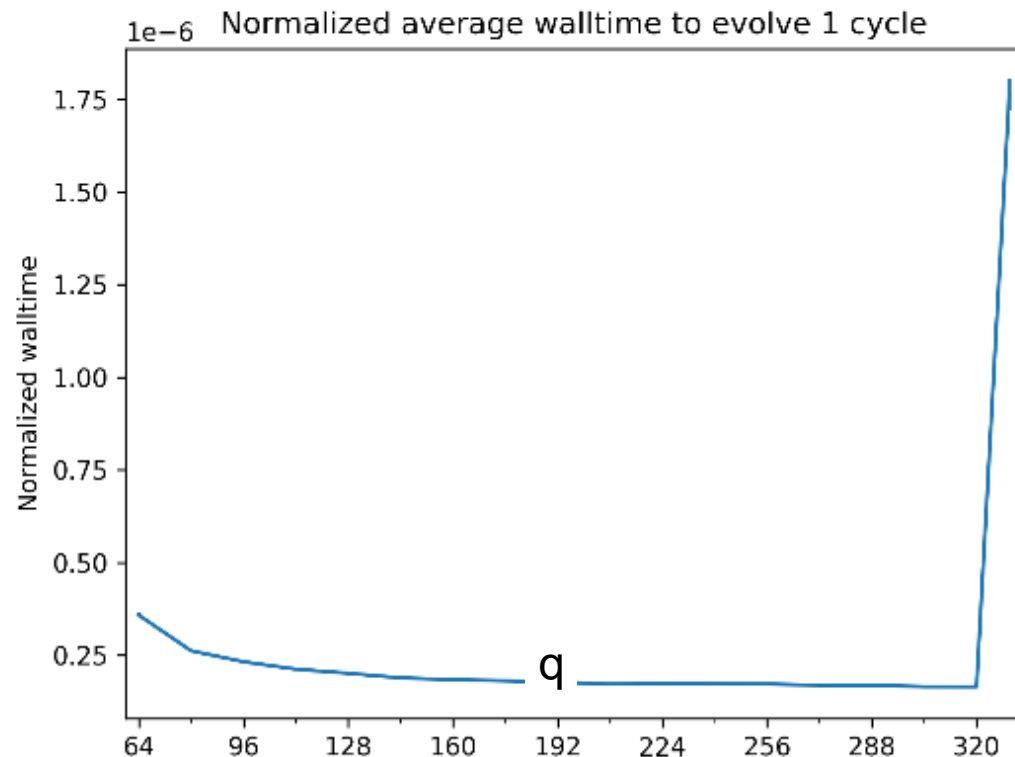
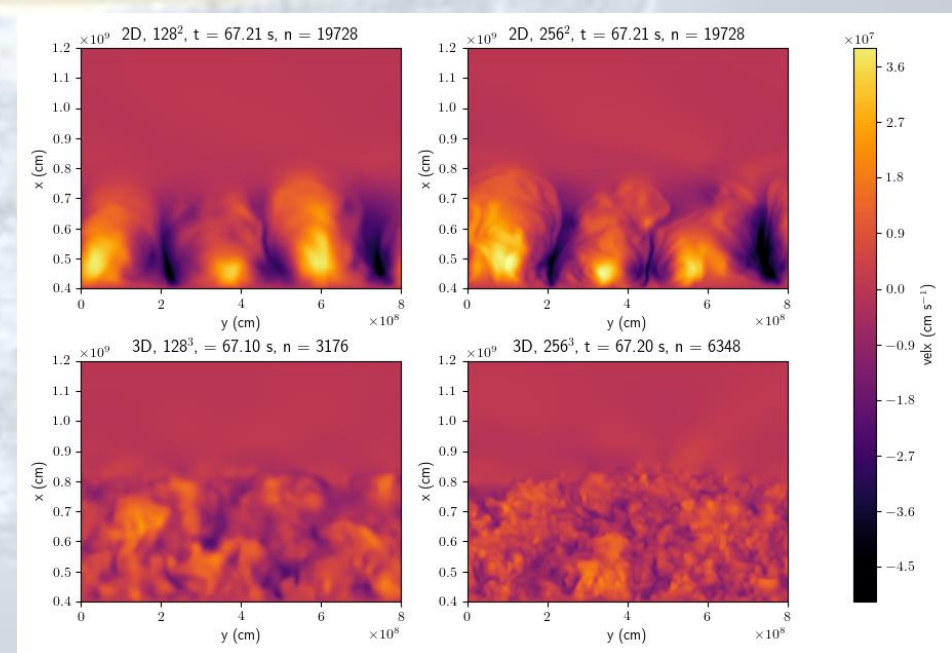
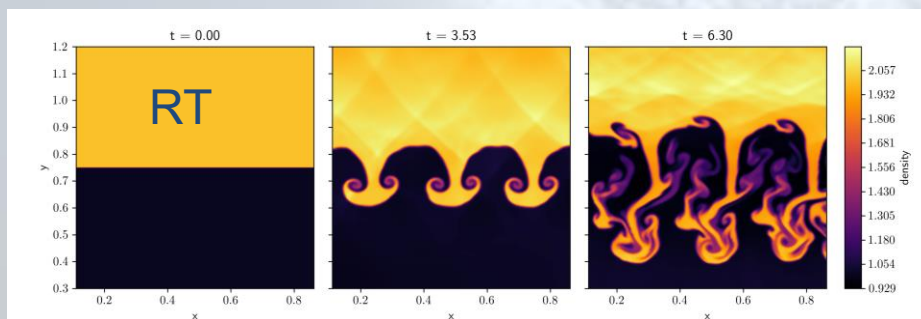
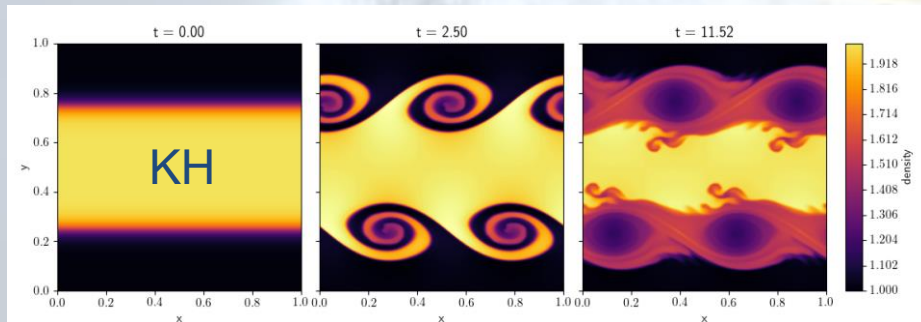


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GPU Porting: Get Set, Ready, ... Not quite!

- Porting and general code development strategy:
 - Version control: <https://gitlab.com/prompi/prompi>
 - Code refactoring & documentation
 - Test suite: simple v-field, KH, RT, turbulent flow
 - Continuous integration (CI) includes simple tests



GPU Porting: Where to Start?

<https://gitlab.com/prompi/prompi>

Physics/task	Code folder within “simulator”
Main programme	PROGRAM
Hydrodynamics/solver	HYDRO
MPI communications	mpi
Equation of state	EOS
Nuclear reactions/neutrinos	BURN
Opacities	OPACITY
...	...

iterative process

How can we go further/faster with GPUs?

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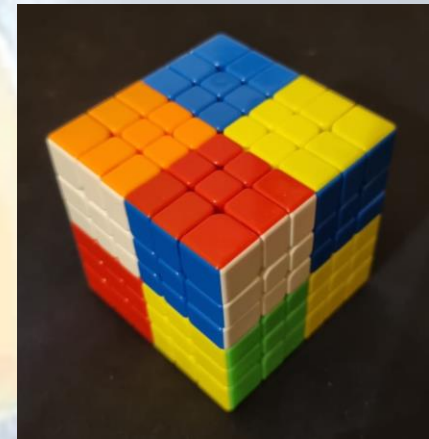
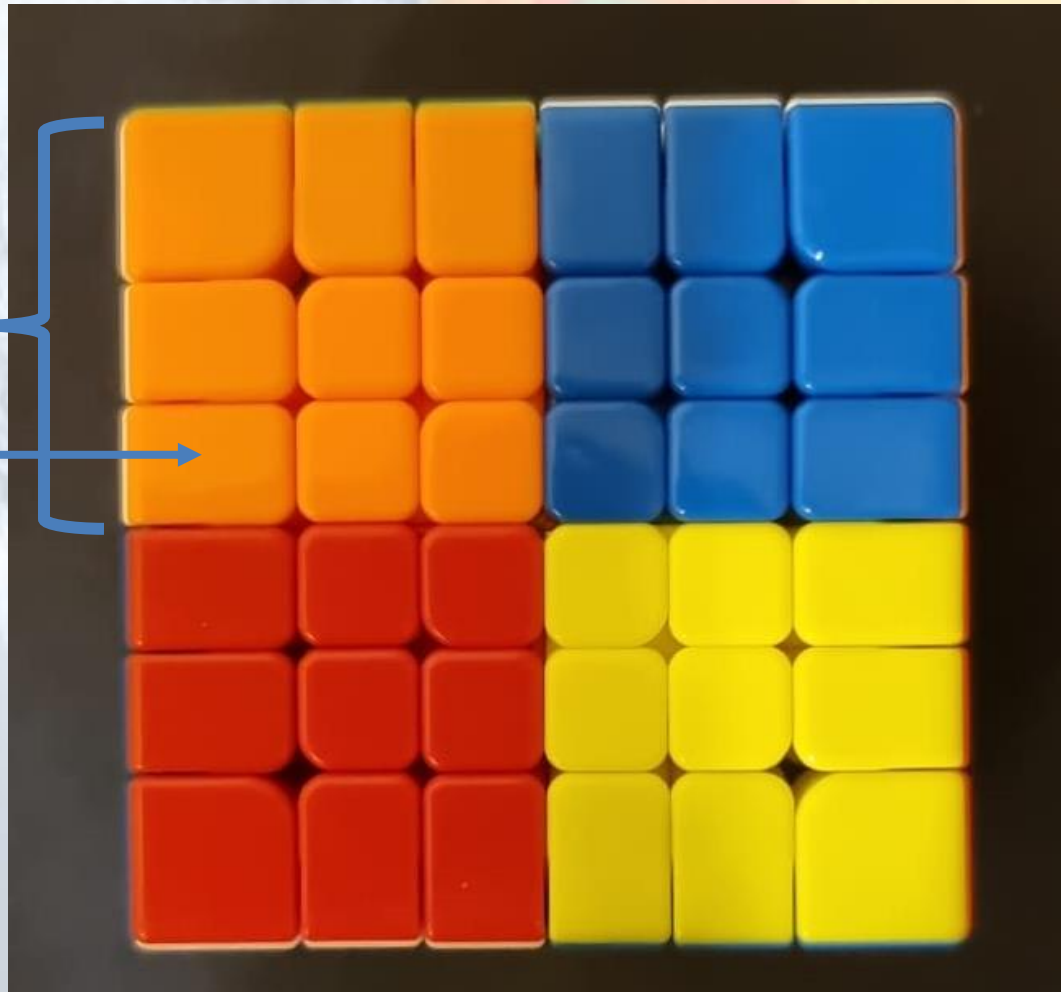


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